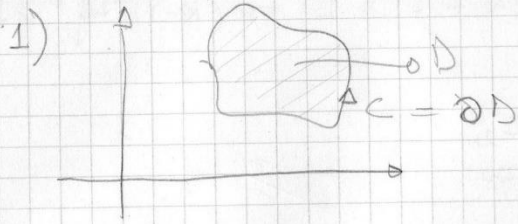


EJERCICIOS RESUELTOS TP 11



$$\frac{1}{k} \int_{\partial D^+} \vec{f} \cdot d\vec{s} = \frac{1}{k} \iint_D (q'_x - p'_y) dx dy =$$

x teorema de Green

$$= \frac{1}{k} \iint_D k dx dy = \frac{k}{k} \iint_D dx dy = \text{área } D$$

• Sea $q'_x - p'_y = 1$, luego:

$$\left. \begin{array}{l} i) \quad q'_x = 1 \Rightarrow q(x,y) = x \\ \quad \quad p'_y = 0 \Rightarrow p(x,y) = 0 \end{array} \right\} \Rightarrow \vec{f} = (p, q) = (0, x)$$

Por lo tanto $\rightarrow \text{área } D = \int_{\partial D^+} (0, x) (dx, dy) =$

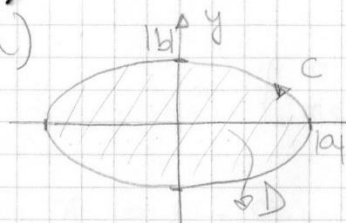
$$= \int_{\partial D^+} x dy$$

$$\left. \begin{array}{l} ii) \quad q'_x = 0 \Rightarrow q(x,y) = 0 \\ \quad \quad -p'_y = 1 \Rightarrow p(x,y) = -y \end{array} \right\} \Rightarrow \vec{f} = (p, q) = (-y, 0)$$

Por lo tanto $\rightarrow \text{área } D = \int_{\partial D^+} (-y, 0) (dx, dy) = \int_{\partial D^+} -y dx$

1)

a)



$$\begin{cases} \frac{x}{a} = \cos t \\ \frac{y}{b} = \sin t \end{cases} \Rightarrow \begin{cases} x = a \cos t \\ y = b \sin t \end{cases}$$

$$t \in [0, 2\pi)$$

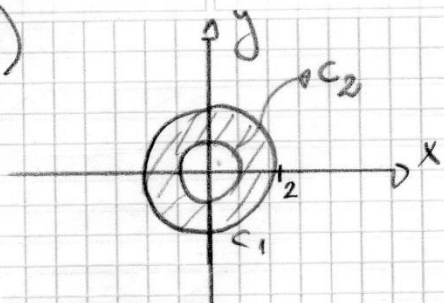
$$\downarrow \\ dy = b \cos t \, dt$$

$$\text{Area } D = \oint_C x \, dy = \int_0^{2\pi} (a \cos t) (b \cos t) \, dt = \int_0^{2\pi} ab \cos^2 t \, dt =$$

$$= ab \left[\frac{1}{2} t + \frac{1}{4} \sin 2t \right]_0^{2\pi} = ab \left[\frac{1}{2} 2\pi + \frac{1}{4} \sin 4\pi - 0 \right] =$$

$$= ab\pi$$

1) b)



$$C_1 \rightarrow (2 \cos t, 2 \sin t) \\ t \in [0, 2\pi)$$

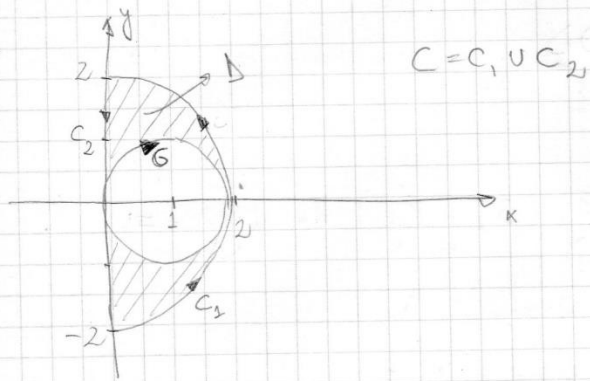
$$C_2 \rightarrow (\cos t, \sin t) \\ t \in [0, 2\pi)$$

$$\text{Area } D = \oint_{C_1} x \, dy - \oint_{C_2} x \, dy = \int_0^{2\pi} 2 \cos t \cdot 2 \cos t \, dt - \int_0^{2\pi} \cos t \cdot \cos t \, dt = \int_0^{2\pi} 3 \cos^2 t \, dt =$$

$$= 3 \left[\frac{1}{2} t + \frac{1}{4} \sin 2t \right]_0^{2\pi} = 3 \left[\frac{1}{2} \cdot 2\pi - 0 \right] = 3\pi$$

c) $x^2 + y^2 \leq 4$; $x^2 + y^2 \geq 2x$; $x \geq 0$

$$x^2 + y^2 - 2x \geq 0 \Rightarrow x^2 + y^2 - 2x + 1 \geq 1 \Rightarrow (x-1)^2 + y^2 \geq 1$$



$$C_1 \rightarrow \begin{cases} x = 1 + \cos t \\ y = \sin t \end{cases} \quad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$$

$$C_2 \rightarrow \begin{cases} x = 0 \\ y = t \end{cases} \quad 2 \geq t \geq -2$$

$$G \rightarrow \begin{cases} x-1 = \cos t \\ y = \sin t \end{cases} \Rightarrow \begin{cases} x = \cos t + 1 \\ y = \sin t \end{cases} \quad (2\pi \geq t \geq 0)$$

$$\text{Area } D = \int_C x \, dy + \int_G x \, dy = \int_{C_1} x \, dy + \int_{C_2} x \, dy + \int_G x \, dy =$$

$$= \int_{-\pi/2}^{\pi/2} 2 \cos t \cdot 2 \cos t \, dt + \int_{-2}^2 0 \cdot dt + \int_{2\pi}^0 (\cos t + 1) \cos t \, dt =$$

$$= 4 \left[\frac{1}{2} t + \frac{1}{4} \sin 2t \right]_{-\pi/2}^{\pi/2} + \int_{2\pi}^0 (\cos^2 t + \cos t) \, dt =$$

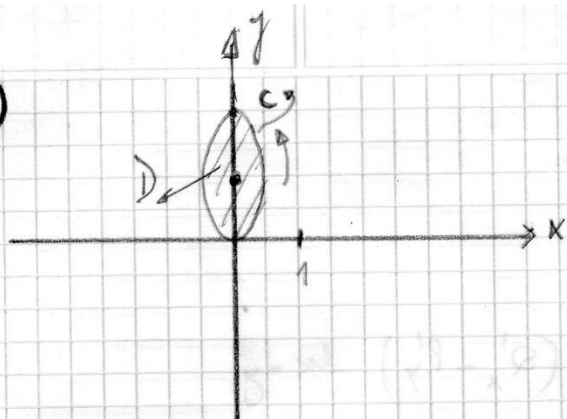
$$= 4 \left[\frac{1}{2} \frac{\pi}{2} + \frac{1}{4} \sin \frac{\pi}{2} \right] - 4 \left[\frac{1}{2} \left(-\frac{\pi}{2} \right) + \frac{1}{4} \sin \left(-\frac{\pi}{2} \right) \right] +$$

$$+ \left[\frac{1}{2} t + \frac{1}{4} \sin 2t \right]_{2\pi}^0 + \left[+ \sin t \right]_{2\pi}^0 = \pi$$

$$2) \text{ Area } D = \iint_D (Q'_x - P'_y) dx dy \quad \begin{array}{l} Q'_x = 1 \Rightarrow Q = x \\ P'_y = 0 \Rightarrow P = 0 \end{array}$$

$$\begin{aligned} \text{Area } D &= \oint_C (0, x) ds = \int_0^1 (0, u - u^2) (1 - 2u, 1 - 4u^3) du = \\ &= \int_0^1 (u - u^2)(1 - 4u^3) du = \int_0^1 (u - u^2 - 4u^4 + 4u^5) du = \\ &= \left[\frac{u^2}{2} - \frac{u^3}{3} - \frac{4u^5}{5} + \frac{4u^6}{6} \right]_0^1 = \frac{1}{2} - \frac{1}{3} - \frac{4}{5} + \frac{4}{6} = \frac{1}{30} \end{aligned}$$

3)



$$\begin{cases} 2x = \rho \cos \theta \\ y - 1 = \rho \sin \theta \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} x = \frac{\rho \cos \theta}{2} \\ y = \rho \sin \theta + 1 \end{cases} \quad \begin{matrix} \theta \in [0, 2\pi) \\ \rho \in [0, 1] \end{matrix}$$

$$4x^2 + (y-1)^2 = 1 \Rightarrow \frac{x^2}{\frac{1}{4}} + (y-1)^2 = 1$$

$$\oint_{C^+} (x^2 + y^2, 3xy + \ln(y^2 + 1)) (dx, dy) =$$

$$= \iint_D (3y - 2y) dx dy = \iint_D y dx dy = \int_0^{2\pi} d\theta \int_0^1 \rho d\rho.$$

$$\bullet (\rho \sin \theta + 1) |J| = (*)$$

$$J = \frac{\partial(x, y)}{\partial(\rho, \theta)} = \begin{vmatrix} \frac{\cos \theta}{2} & -\frac{\rho \sin \theta}{2} \\ \sin \theta & \rho \cos \theta \end{vmatrix} = \frac{\rho}{2} \cos^2 \theta +$$

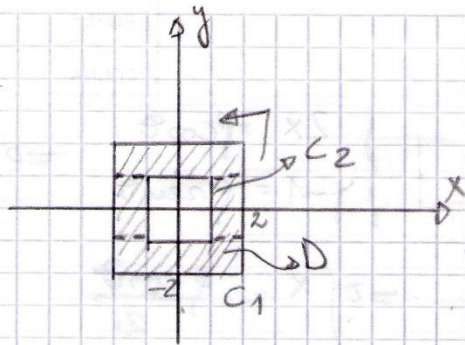
$$+ \frac{\rho}{2} \sin^2 \theta = \frac{\rho}{2}$$

$$(*) = \int_0^{2\pi} d\theta \int_0^1 (\rho \sin \theta + 1) \frac{\rho}{2} d\rho = \int_0^{2\pi} d\theta \left[\frac{\rho^3}{6} \sin \theta + \right.$$

$$\left. + \frac{\rho^2}{4} \right]_0^1 = \int_0^{2\pi} d\theta \left[\frac{\sin \theta}{6} + \frac{1}{4} \right] = \left[-\frac{\cos \theta}{6} + \frac{\theta}{4} \right]_0^{2\pi} =$$

$$= -\frac{1}{6} + \frac{2\pi}{4} + \frac{1}{6} = \frac{\pi}{2}$$

4)



$$\oint_{C_1^+} \vec{F} \cdot d\vec{g} - \oint_{C_2^+} \vec{F} \cdot d\vec{g} = \iint_D (Q'_x - P'_y) dx dy$$

$$C_1 \begin{cases} (x, -2) & -2 \leq x \leq 2 \\ (2, y) & -2 \leq y \leq 2 \\ (x, 2) & 2 \geq x \geq -2 \\ (-2, y) & 2 \geq y \geq -2 \end{cases}$$

$$C_2 \begin{cases} (x, -1) & -1 \leq x \leq 1 \\ (1, y) & -1 \leq y \leq 1 \\ (x, 1) & 1 \geq x \geq -1 \\ (-1, y) & 1 \geq y \geq -1 \end{cases}$$

$$\oint_{C_1^+} \vec{F} \cdot d\vec{g} = \int_{-2}^2 (x^2 y, y^2) (1, 0) dx + \int_{-2}^2 (x^2 y, y^2) (0, 1) dy +$$

$$+ \int_2^{-2} (x^2 y, y^2) (1, 0) dx + \int_2^{-2} (x^2 y, y^2) (0, 1) dy =$$

$$= \int_{-2}^2 x^2 \overbrace{y}^{-2} dx + \int_{-2}^2 \cancel{y^2} dy + \int_2^{-2} x^2 \overbrace{y}^2 dx + \int_2^{-2} \cancel{y^2} dy =$$

$$= -2 \int_{-2}^2 x^2 dx - 2 \int_{-2}^2 x^2 dx = -4 \left[\frac{x^3}{3} \Big|_{-2}^2 \right] =$$

$$= -4 \left(\frac{8}{3} - \left(-\frac{8}{3} \right) \right) = -4 \cdot \frac{16}{3} = -\frac{64}{3}$$

$$\oint_{C_2^+} \vec{F} \cdot d\vec{y} = \int_{-1}^1 (x^2 y, y^2) (1, 0) dx + \int_{-2}^2 (x^2 y, y^2) (0, 1) dy +$$

$$+ \int_{-1}^1 x^2 \frac{-1}{y} dx + \int_{-1}^1 y^2 \frac{1}{y} dy + \int_1^{-1} x^2 \frac{1}{y} dx + \int_2^{-2} y^2 \frac{1}{y} dy =$$

$$= - \int_{-1}^1 x^2 dx - \int_{-1}^1 x^2 dx = -2 \left[\frac{x^3}{3} \Big|_{-1}^1 \right] = -2 \left(\frac{1}{3} + \frac{1}{3} \right) =$$

$$= -2 \cdot \frac{2}{3} = -\frac{4}{3}$$

$$\oint_{C_1^+} - \oint_{C_2^+} = -\frac{64}{3} - \left(-\frac{4}{3} \right) = -\frac{60}{3} = -20$$

$$\iint_D (Q'_x - P'_y) dx dy = \iint_D (0 - x^2) dx dy = \int_{-2}^2 dx \int_{-2}^{-1} -x^2 dy +$$

$$+ \int_{-2}^2 dx \int_1^2 -x^2 dy + \int_{-2}^{-1} dx \int_{-1}^1 -x^2 dy + \int_1^2 dx \int_{-1}^1 -x^2 dy =$$

$$= \int_{-2}^2 -x^2 dx + \int_{-2}^2 -x^2 dx + \int_{-2}^{-1} x^2 dx + \int_1^2 -2x^2 dx =$$

$$= -2 \left[\frac{x^3}{3} \Big|_{-2}^2 \right] - 2 \left[\frac{x^3}{3} \Big|_{-2}^{-1} \right] - 2 \left[\frac{x^3}{3} \Big|_1^2 \right] =$$

$$= -2 \left[\frac{8}{3} - \left(-\frac{8}{3} \right) \right] - 2 \left(-\frac{1}{3} - \left(-\frac{8}{3} \right) \right) - 2 \left(\frac{8}{3} - \frac{1}{3} \right) =$$

$$= -2 \cdot \frac{16}{3} - 2 \cdot \frac{7}{3} - 2 \cdot \frac{7}{3} = -2 \cdot \left(\frac{14}{3} + \frac{16}{3} \right) = -\frac{60}{3} = -20$$

the formula

$$\begin{aligned}
 \iint_D (Q'_x - P'_y) dx dy &= \iint_{D_1} -x^2 dx dy - \iint_{D_2} -x^2 dx dy = \\
 &= \int_{-2}^2 dx \int_{-2}^2 -x^2 dy + \int_{-1}^1 dx \int_{-1}^1 x^2 dy = - \int_{-2}^2 4x^2 dx + \\
 &+ \int_{-1}^1 2x^2 dx = -4 \left[\frac{x^3}{3} \right]_{-2}^2 + 2 \left[\frac{x^3}{3} \right]_{-1}^1 = \\
 &= -4 \left(\frac{8}{3} + \frac{8}{3} \right) + 2 \left(\frac{1}{3} + \frac{1}{3} \right) = -4 \cdot \frac{16}{3} + 2 \cdot \frac{2}{3} = \\
 &= -\frac{64+4}{3} = -\frac{60}{3} = -20
 \end{aligned}$$

$$5) \quad \int_{C_{BA}} \vec{f} \cdot d\vec{s} + \int_{C_{AB}} \vec{f} \cdot d\vec{s} = \oint_{C_{UAB}} \vec{f} \cdot d\vec{s} = \iint_D (Q'_x - P'_y) dx dy = 0$$

$$\begin{aligned}
 \Rightarrow \int_{C_{BA}} \vec{f} \cdot d\vec{s} &= \iint_D (3/x + 2 - 3/x + 1) dx dy - 17 = 3 \iint_D dx dy - \\
 -17 &= 3 \int_a^b dx \int_0^{x^2-x^4} dy - 17 =
 \end{aligned}$$

$$A \rightarrow B \Rightarrow x^2 - x^4 = 0 \Rightarrow x^2(1-x^2) = 0 \Rightarrow x^2(1-x)(1+x) = 0 \Rightarrow$$

$$\Rightarrow x=0 \vee x=1 \vee x=-1 \Rightarrow A=(-1,0)$$

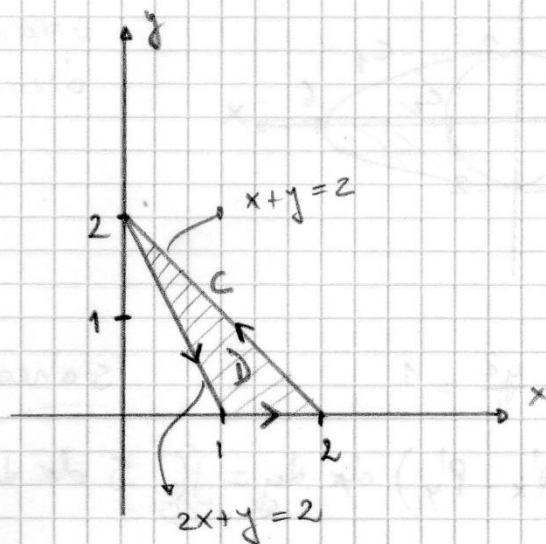
$$B=(1,0)$$

$$= 3 \int_{-1}^1 (x^2 - x^4) dx - 17 = 3 \left[\frac{x^3}{3} - \frac{x^5}{5} \right]_{-1}^1 - 17 =$$

$$= 3 \left[\frac{1}{3} - \frac{1}{5} + \frac{1}{3} - \frac{1}{5} \right] - 17 = 3 \cdot \frac{4}{5} - 17 = -81/5$$

$$\int_{C_{BA}} \vec{f} \cdot d\vec{s} = -\frac{81}{5} \Rightarrow \int_{C_{AB}} \vec{f} \cdot d\vec{s} = \frac{81}{5}$$

6)



$$a) \quad P = 2y - g(x) \\ Q = 5x - h(y)$$

$$\oint_{C^+} \vec{F} \cdot d\vec{g} = \iint_D (Q'_x - P'_y) \, dx \, dy = \iint_D (5 - 2) \, dx \, dy =$$

$$= 3 \int_0^2 dy \int_{\frac{2-y}{2}}^{2-y} dx = 3 \int_0^2 \left[2-y - \frac{2-y}{2} \right] dy =$$

$$= 3 \int_0^2 dy \frac{2-y}{2} = \frac{3}{2} \left[2y - \frac{y^2}{2} \right]_0^2 = \frac{3}{2} \left[4 - \frac{4}{2} \right] =$$

$$= \frac{3}{2} \cdot 2 = 3$$

$$b) \quad P = 2y + g(x-y) \\ Q = 2x - g(x-y)$$

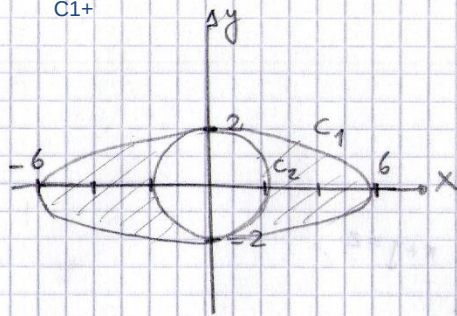
$$\oint_{C^+} \vec{F} \cdot d\vec{g} = \iint_D (Q'_x - P'_y) \, dx \, dy = \iint_D \left[2 - g' - (2 + \overbrace{g'(-1)}^{-g'}) \right] dx \, dy$$

$$= \iint_D [2 - g' - 2 + g'] \, dx \, dy = 0$$

$$\begin{aligned}
 7) \oint_{\partial D^+} \vec{f} \cdot d\vec{s} &= \iint_D (Q'_x - P'_y) dx dy = \iint_D (h'(x) + 1) dx dy \\
 &= \int_{-2}^2 dx \int_{-3}^3 (h'(x) + 1) dy = 6 \int_{-2}^2 (h'(x) + 1) dx = \\
 &= 6 \left[h(x) + x \right]_{-2}^2 = 6 (h(2) + 2 - h(-2) + 2) = \\
 &= 24 + 6h(2) - 6h(-2) = 24 + \cancel{6h(2)} - \underbrace{6h(-2)}_{\text{hes por}} = \\
 &= 24
 \end{aligned}$$

Sea $\vec{f} = (P, Q)$ una función de clase C^1 en $\mathbb{R}^2 - \{(0,0)\}$ tal que $Q'_x - P'_y = 5$, dadas las curvas $C_1: x^2 + 9y^2 = 36$ y $C_2: x^2 + y^2 = 4$. Calcule $\oint_{C_2^+} \vec{f} \cdot d\vec{s}$ sabiendo que $\oint_{C_1^+} \vec{f} \cdot d\vec{s} = 7\pi$

8)



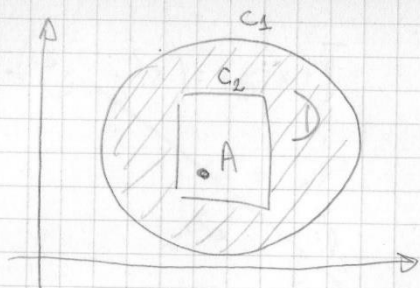
área elipse $\rightarrow ab\pi$
 área círculo $\rightarrow \pi r^2$

$$\begin{aligned}
 x^2 + y^2 &= 4 \\
 x^2 + 9y^2 &= 36 \Rightarrow \frac{x^2}{36} + \frac{y^2}{4} = 1
 \end{aligned}$$

$$\underbrace{\oint_{C_1^+} \vec{f} \cdot d\vec{s}}_{7\pi} - \oint_{C_2^+} \vec{f} \cdot d\vec{s} = \iint_D (Q'_x - P'_y) dx dy = \iint_D 5 dx dy$$

$$\begin{aligned}
 \oint_{C_2^+} \vec{f} \cdot d\vec{s} &= 7\pi - 5(6 \cdot 2 \cdot \pi - \pi \cdot 4) = 7\pi - 5(12\pi - 4\pi) = \\
 &= 7\pi - 40\pi = -33\pi
 \end{aligned}$$

9)



$$\iint_D (2x - y) \, dx \, dy = \int_{C_1^+} \vec{F} \cdot d\vec{s} - \int_{C_2^+} \vec{F} \cdot d\vec{s} \Rightarrow$$

$$\Rightarrow \iint_D 6 \, dx \, dy = \int_{C_1^+} \vec{F} \cdot d\vec{s} - 12 \Rightarrow$$

$$\Rightarrow 6 \iint_D 1 \, dx \, dy + 12 = \int_{C_1^+} \vec{F} \cdot d\vec{s} \Rightarrow$$

$$\Rightarrow 6 \text{ area } D + 12 = \int_{C_1^+} \vec{F} \cdot d\vec{s} \Rightarrow$$

$$\Rightarrow 6 [\pi \cdot 8^2 - 5^2] + 12 = \int_{C_1^+} \vec{F} \cdot d\vec{s} \Rightarrow$$

$$\Rightarrow \int_{C_1^+} \vec{F} \cdot d\vec{s} = 384\pi - 150 + 12 = 384\pi - 138$$

$$10) \vec{f}: \mathbb{R}^2 \rightarrow \mathbb{R}^2 \quad D\vec{f} \text{ continua y simétrica}$$

$$\forall C \text{ en } \mathbb{R}^2 : \oint_C \vec{f} \, d\vec{s}$$

$$\text{Sea } \vec{f}(x,y) = (P(x,y), Q(x,y))$$

$$D\vec{f} = \begin{pmatrix} P'_x & P'_y \\ Q'_x & Q'_y \end{pmatrix} \text{ continua y simétrica}$$

$$Q'_x = P'_y \Rightarrow Q'_x - P'_y = 0$$

$$\oint_C \vec{f} \, d\vec{s} \stackrel{\text{pot. Green}}{=} \iint_D (Q'_x - P'_y) \, dx \, dy = \iint_D 0 \, dx \, dy = 0$$

Para aplicar T. Green, las hipótesis sobre C son:

C una curva cerrada, simple, suave a trozos

C curva frontera de un recinto simple D

$$D \subset \text{Dom } \vec{f} \quad (\because C \subset \text{Dom } \vec{f})$$

$$11) \vec{f}(x,y,z) = (9x^2 + 2y + y^2, 2x + 2xy)$$

$$\frac{\partial(9x^2 + 2y + y^2)}{\partial y} = 2 + 2y$$

$$\frac{\partial(2x + 2xy)}{\partial x} = 2 + 2y$$

$\text{Dom } \vec{f} = \mathbb{R}^3$ simplemente
conexo

$\vec{f}$ admite función potencial

$$\begin{cases} \phi'_x = 9x^2 + 2y + y^2 \\ \phi'_y = 2x + 2xy \end{cases}$$

$$\phi = \int (9x^2 + 2y + y^2) dx = 3x^3 + 2yx + y^2x + g(y)$$

$$\begin{cases} \phi'_y = 2x + 2xy + g'(y) \\ \phi'_y = 2x + 2xy \end{cases} \Rightarrow g'(y) = 0 \Rightarrow g(y) = C$$

$$\phi(x,y) = 3x^3 + 2yx + y^2x + C$$

$$\phi(0,0) = 2 \Rightarrow C = 2 \quad \therefore \phi(x,y) = 3x^3 + 2yx + y^2x + 2$$

Cálculo extremo $\begin{cases} \phi'_x = 9x^2 + 2y + y^2 = 0 & (1) \\ \phi'_y = 2x + 2xy = 0 & (2) \end{cases}$

De (2): $2x(1+y) = 0 \Rightarrow x = 0 \vee y = -1$

$x = 0$ en (1) $\rightarrow 2y + y^2 = 0 \Rightarrow y(2+y) = 0 \rightarrow \begin{matrix} y = 0 \\ y = -2 \end{matrix}$

$P_1 = (0,0) \quad P_2 = (0,-2)$

$y = -1$ en (1) $\rightarrow 9x^2 - 2 + 1 = 0 \Rightarrow x^2 = 1/9 \rightarrow \begin{matrix} x = 1/3 \\ x = -1/3 \end{matrix}$

$P_3 = (1/3, -1) \quad P_4 = (-1/3, -1)$

$$H = \begin{vmatrix} 18x & 2+2y \\ 2+2y & 2x \end{vmatrix}$$

$$H(0,0) = \begin{vmatrix} 0 & 2 \\ 2 & 0 \end{vmatrix} < 0 \rightarrow (0,0) \text{ pto silla}$$

$$H(0,-2) = \begin{vmatrix} 0 & -2 \\ -2 & 0 \end{vmatrix} < 0 \rightarrow (0,-2) \text{ pto silla}$$

$$H\left(\frac{1}{3}, -1\right) = \begin{vmatrix} \frac{18}{3} & 0 \\ 0 & \frac{2}{3} \end{vmatrix} > 0 \rightarrow \text{extremo}$$

$$\phi''_{xx}\left(\frac{1}{3}, -1\right) = \frac{18}{3} > 0 \Rightarrow \phi\left(\frac{1}{3}, -1\right) \text{ es m\u00edn local}$$

$$H\left(-\frac{1}{3}, -1\right) = \begin{vmatrix} -\frac{18}{3} & 0 \\ 0 & -\frac{2}{3} \end{vmatrix} > 0 \rightarrow \text{extremo}$$

$$\phi''_{xx}\left(-\frac{1}{3}, -1\right) = -\frac{18}{3} < 0 \Rightarrow \phi\left(-\frac{1}{3}, -1\right) \text{ es m\u00e1x local}$$

$$\phi\left(-\frac{1}{3}, -1\right) = \frac{20}{9} \text{ es m\u00e1x local}$$

$$\phi\left(\frac{1}{3}, -1\right) = \frac{16}{9} \text{ es m\u00edn local}$$

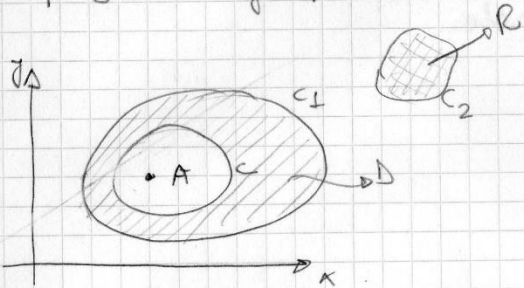
$$12) \oint_{C_3^+} \bar{f} \, d\bar{z} - \int_{C_1^+} \bar{f} \, d\bar{z} - \int_{C_2^+} \bar{f} \, d\bar{z} = 0 \Rightarrow 0$$

$$\Rightarrow \oint_{C_3^+} \bar{f} \, d\bar{z} - 12\pi + \oint_{C_2^-} \bar{f} \, d\bar{z} = 0 \Rightarrow 0$$

$$\Rightarrow \oint_{C_3^+} \bar{f} \, d\bar{z} = 12\pi - \oint_{C_2^-} \bar{f} \, d\bar{z} \Rightarrow \int_{C_3^+} \bar{f} \, d\bar{z} = 12\pi - 16\pi = -4\pi$$

13) $\vec{f}$ continua y simétrica en $\mathbb{R}^2 - \{A\}$ $\therefore$

$$D\vec{f} = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} \end{pmatrix} \in C^1 \text{ en } \mathbb{R}^2 - \{A\} \quad \wedge \quad \frac{\partial f_1}{\partial y} = \frac{\partial f_2}{\partial x} \text{ en } \mathbb{R}^2 - \{A\}$$



Aplicamos el T. de Green con la f $\vec{f}$ al recinto Δ

$$\vec{f} = (f_1, f_2) \rightarrow P = f_1, Q = f_2$$

$$\iint_{\Delta} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) dx dy = \int_{C_1^+} \vec{f} \cdot d\vec{s} + \int_{C^-} \vec{f} \cdot d\vec{s}$$

$$\iint_{\Delta} 0 \, dx \, dy \quad (\text{por H}) = \int_{C_1^+} \vec{f} \cdot d\vec{s} - \int_{C^+} \vec{f} \cdot d\vec{s} \Rightarrow$$

$$\Rightarrow \int_{C_1^+} \vec{f} \cdot d\vec{s} - \int_{C^+} \vec{f} \cdot d\vec{s} = 0 \Rightarrow \int_{C_1^+} \vec{f} \cdot d\vec{s} = \int_{C^+} \vec{f} \cdot d\vec{s}$$

Siendo C_1 y C las curvas cualquiera que rodean al pto A , el resultado de la circulación de cualquier curva cerrada que encierre al pto A es el mismo valor-

$$\oint_{C^+} \vec{f} \cdot d\vec{s} = 0 \Rightarrow \exists \phi / \nabla \phi = \vec{f}$$

$\oint_{C^+} \vec{f} \cdot d\vec{s} = 0 \Rightarrow \oint_{C^+} \vec{f} \cdot d\vec{s} = 0$ y el valor de cualquier integral curvilínea sobre una curva cerrada que rodea a $\bar{A}$ es cero-

Si C no rodea a A (p.ej C_2), tenemos que se podrá aplicar el T. de Green al recinto R

$$\oint_{C_2} \vec{f} \cdot d\vec{s} = \iint_R \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) dx dy = \iint_R 0 \, dx \, dy = 0$$

Por lo tanto, para cualquier curva cerrada (que encierre o no al pto A), $\oint_C \vec{f} \cdot d\vec{s} = 0 \quad \forall C$ cerrada. Por lo tanto $\vec{f}$ es conservativo en $\mathbb{R}^2 - \{A\}$ y es un campo de gradientes. Luego existe $\phi / \vec{f} = \nabla \phi$ en $\mathbb{R}^2 - \{A\}$

$$\Rightarrow \exists \phi / \vec{f} = \nabla \phi \text{ en } \mathbb{R}^2 - \{A\} \Rightarrow \int_{C^+} \vec{f} \cdot d\vec{s} = 0$$

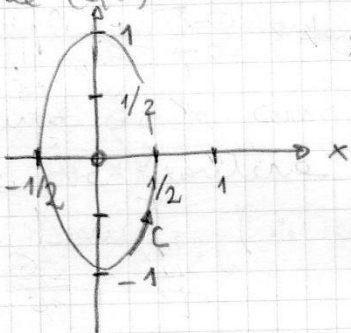
Si $\vec{f} = \nabla \phi$ en $\mathbb{R}^2 - \{A\}$, luego $\vec{f}$ es conservativo en $\mathbb{R}^2 - \{A\}$, y por lo tanto $\int_{C^+} \vec{f} \cdot d\vec{s} = 0$ por ser C^+ una curva cerrada.

$$14) \bar{f}: \mathbb{R}^2 - \{0\} \rightarrow \mathbb{R}^2 / \bar{f}(xy) = \left(\underbrace{x + \frac{y}{4x^2+y^2}}_P, \underbrace{y - \frac{x}{4x^2+y^2}}_{Q'} \right)$$

$$P'_y = \frac{4x^2+y^2 - 2y^2}{(4x^2+y^2)^2} = \frac{4x^2-y^2}{(4x^2+y^2)^2}$$

$$Q'_x = \frac{-(4x^2+y^2) + 8x^2}{(4x^2+y^2)^2} = \frac{4x^2-y^2}{(4x^2+y^2)^2}$$

Siendo $\text{Dom } \bar{f} = \mathbb{R}^2 - \{0\}$ un conj. que no es simplemente conexo, elijo una curva que encierre al $(0,0)$



$$4x^2 + y^2 = 1$$

$$\frac{x^2}{\frac{1}{4}} + \frac{y^2}{1} = 1$$

$$C: \bar{g}(t) = \left(\frac{1}{2} \cos t, \sin t \right)$$

$$0 \leq t \leq 2\pi$$

$$\oint_C \bar{f} \, ds = \int_0^{2\pi} \left(\frac{1}{2} \cos t + \frac{\sin t}{4 \underbrace{\frac{1}{4} \cos^2 t + \sin^2 t}_1} \right) \sin t - \underbrace{\frac{1}{2} \cos t}_{\frac{4}{4} \cos^2 t + \sin^2 t} \cos t \, dt$$

$$\cdot \left(-\frac{1}{2} \sin t, \cos t \right) dt =$$

$$= \int_0^{2\pi} \left[\left(\frac{1}{2} \cos t + \sin t \right) \left(-\frac{1}{2} \sin t \right) + \left(\sin t - \frac{1}{2} \cos t \right) \cos t \right] dt =$$

$$= \int_0^{2\pi} \left(-\frac{1}{4} \sin t \cos t - \frac{1}{2} \sin^2 t + \sin t \cos t - \frac{1}{2} \cos^2 t \right) dt =$$

$$= \int_0^{2\pi} \frac{3}{4} \sin t \cos t \, dt - \frac{1}{2} \int_0^{2\pi} dt = \left[\frac{3}{4} \frac{\sin^2 t}{2} \right]_0^{2\pi} - \frac{1}{2} \cdot 2\pi =$$

$$= \frac{3}{4} \left[\frac{\sin^2 2\pi}{2} - \frac{\sin^2 0}{2} \right] - \pi = -\pi \neq 0$$

Si $\bar{f}$ admite f. potencial, debería ser $\oint_C \bar{f} \, ds = 0$

4c. Sin embargo hay una curva donde la integral es $\neq 0$.

Por lo tanto no admite f potencial

15) Sea $\vec{f} / D\vec{f}$ continua y simétrica en $\mathbb{R}^3 - \{\vec{A}\}$

i) En el espacio, $\mathbb{R}^3 - \{\vec{A}\}$ es un conj. abierto y simplemente conexo

ii) $D\vec{f}$ continua y simétrica en $\mathbb{R}^3 - \{\vec{A}\}$ y

Por además $\text{rot}(\vec{f}) = \vec{0}$

Por lo tanto

$$\oint_{\Sigma} \text{rot} \vec{f} \cdot \vec{n} \, d\sigma = 0 \quad \forall \text{ sup. } \Sigma \text{ que cumple las condiciones del T. de Stokes}$$

Aplicando dicho teorema:

$$\oint_{\Sigma} \text{rot} \vec{f} \cdot \vec{n} \, d\sigma = \oint_C \vec{f} \cdot d\vec{g} = 0$$

Sea $\oint_C \vec{f} \cdot d\vec{g} = 0 \quad \forall \text{ curva } C \text{ cerrada que es borde de la sup. } \Sigma$

Si $\oint_C \vec{f} \cdot d\vec{g} = 0, \forall \text{ curva } C$, entonces $\vec{f}$ es un

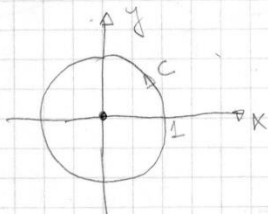
campo conservativo y existe función potencial de él.

Luego: $\exists \phi / \phi: \mathbb{R}^3 - \{\vec{A}\} \rightarrow \mathbb{R} \wedge \vec{f} = \nabla \phi$

16) $\vec{E}(x, y) = kq \frac{(x, y)}{(x^2 + y^2)^{3/2}}$

$\vec{E}$ cumple con las condiciones pedidas en el ítem 3) (*)

Dom $\vec{E} = \mathbb{R}^2 - \{0\}$



$C: \vec{g}(t) = (\cos t, \sin t), 0 \leq t \leq 2\pi$

Se $\int_C \vec{E} \cdot d\vec{s} = 0$, por (*), $\vec{E}$ admite potencial

$$\int_{C^+} \vec{E} \cdot d\vec{s} = \int_{C^+} \frac{kq(x, y)}{(x^2 + y^2)^3} d\vec{g} = \int_0^{2\pi} \frac{kq(\cos t, \sin t)}{(\sqrt{\cos^2 t + \sin^2 t})^3} \cdot (-\sin t, \cos t) dt = 0$$

$\cdot (-\sin t, \cos t) dt = \int_0^{2\pi} kq (-\cos t \sin t + \cos t \sin t) dt = 0$

Análisis $\vec{E}$: es continua en $\mathbb{R}^2 - \{0\}$ (compuesta de continuas) simétrica en $\mathbb{R}^2 - \{0\}$

$$\vec{D}\vec{E} = \begin{bmatrix} \frac{(x^2 + y^2)^{3/2} - \frac{3}{2}(x^2 + y^2)^{1/2} \cdot x \cdot 2x}{(x^2 + y^2)^3} & \frac{0 - \frac{3}{2}(x^2 + y^2)^{1/2} \cdot x \cdot 2y}{(x^2 + y^2)^3} \\ \frac{0 - \frac{3}{2}(x^2 + y^2)^{1/2} \cdot y \cdot 2x}{(x^2 + y^2)^3} & \frac{(x^2 + y^2)^{3/2} - \frac{3}{2}(x^2 + y^2)^{1/2} \cdot y \cdot 2y}{(x^2 + y^2)^3} \end{bmatrix}$$

$P = \frac{kq x}{(x^2 + y^2)^{3/2}}$

$Q = \frac{kq y}{(x^2 + y^2)^{3/2}}$

$$\phi = \int P dx = kq \int \frac{x}{(x^2+y^2)^{3/2}} dx = kq \int \frac{x}{(\sqrt{x^2+y^2})^3} dx =$$

$$= kq \left(-\frac{1}{\sqrt{x^2+y^2}} \right) + g(y) = \frac{-kq}{\sqrt{x^2+y^2}} + g(y)$$

$$\left. \begin{aligned} \phi'_y &= +kq \left(+\frac{1}{2} (x^2+y^2)^{-3/2} \cdot 2y \right) + g'(y) = \frac{kq y}{(\sqrt{x^2+y^2})^3} + g'(y) \\ \phi'_y &= \frac{kq y}{(\sqrt{x^2+y^2})^3} \end{aligned} \right\} \Rightarrow$$

$$\Rightarrow g'(y) = 0 \Rightarrow g(y) = C$$

$$\phi(x,y) = \frac{-kq}{\sqrt{x^2+y^2}} + C \Rightarrow \boxed{U = -\phi = \frac{kq}{\sqrt{x^2+y^2}} + C_1}$$

$$\text{So } U(\infty, \infty) = 0 \Rightarrow C_1 = 0$$

FECHA _____

$$17) \vec{E}(\vec{r}) = kq \frac{\vec{r}}{r^3} = \frac{kq(x, y, z)}{(\sqrt{x^2 + y^2 + z^2})^3} =$$

$$= \frac{kq(x, y, z)}{(x^2 + y^2 + z^2)^{3/2}}$$

$\text{Dom } \vec{E} = \mathbb{R}^3 - \{0\}$ es simplemente conexo

$$\text{rot } \vec{E} = kq \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \underline{x} & \underline{y} & \underline{z} \end{vmatrix} =$$

$$= kq \left[\hat{i} \left(-\frac{3}{2} \frac{(x^2 + y^2 + z^2)^{1/2} \cdot 2yz}{(x^2 + y^2 + z^2)^3} - \frac{3}{2} \frac{(x^2 + y^2 + z^2)^{1/2} \cdot 2yz}{(x^2 + y^2 + z^2)^3} \right) - \right.$$

$$\left. - \hat{j} \left(-\frac{3}{2} \frac{(x^2 + y^2 + z^2)^{1/2} \cdot 2xz}{(x^2 + y^2 + z^2)^3} - \frac{3}{2} \frac{(x^2 + y^2 + z^2)^{1/2} \cdot 2zx}{(x^2 + y^2 + z^2)^3} \right) + \right.$$

$$\left. + \hat{k} \left(-\frac{3}{2} \frac{(x^2 + y^2 + z^2)^{1/2} \cdot 2xy}{(x^2 + y^2 + z^2)^3} - \frac{3}{2} \frac{(x^2 + y^2 + z^2)^{1/2} \cdot 2yx}{(x^2 + y^2 + z^2)^3} \right) \right] =$$

$= kq(0, 0, 0) \Rightarrow$ según ej anterior $\exists \phi$ potencial

$$\phi_x = \frac{kq x}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\phi_y = \frac{kq y}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\phi_z = \frac{kq z}{(x^2 + y^2 + z^2)^{3/2}}$$

$$\phi = \int \frac{kq x}{(x^2 + y^2 + z^2)^{3/2}} dx = kq \int \frac{x}{\sqrt{(x^2 + y^2 + z^2)^3}} dx = -\frac{kq}{\sqrt{(x^2 + y^2 + z^2)}} +$$

$$+ \phi(y, z)$$

$$\phi'_y = -kq \left(-\frac{1}{2} \frac{2y}{\sqrt{(x^2+y^2+z^2)^3}} \right) + \lambda'_y = \frac{kq y}{\sqrt{(x^2+y^2+z^2)^3}} \Rightarrow \lambda'_y = 0 \Rightarrow$$

$$\Rightarrow \lambda(y, z) = \lambda(z)$$

$$\phi = \frac{-kq}{\sqrt{x^2+y^2+z^2}} + \lambda(z)$$

$$\phi'_z = -kq \left(-\frac{1}{2} (x^2+y^2+z^2)^{-3/2} \cdot 2z \right) + \lambda'_z = \frac{kq z}{\sqrt{(x^2+y^2+z^2)^3}} \Rightarrow \lambda'_z = 0 \Rightarrow$$

$$\Rightarrow \lambda(z) = C$$

$$\phi(x, y, z) = \frac{-kq}{\sqrt{x^2+y^2+z^2}} + C$$

$$\vec{E} = -\nabla U \Rightarrow U = \frac{kq}{\sqrt{x^2+y^2+z^2}} + C_1$$

$$\text{Se } U(\infty, \infty, \infty) = 0 \Rightarrow C_1 = 0$$

18) a) Dom $\bar{f} = \mathbb{R}^2 - \{(1, 0)\}$

$$\frac{\partial f_1}{\partial y} = \frac{(x-1)^2 + y^2 - 2y^2}{((x-1)^2 + y^2)^2} = \frac{(x-1)^2 - y^2}{((x-1)^2 + y^2)^2}$$

$$\frac{\partial f_2}{\partial x} = \frac{-(x-1)^2 - y^2 - 2(x-1)(1-x)}{((x-1)^2 + y^2)^2} = \frac{(x-1)^2 - y^2}{((x-1)^2 + y^2)^2}$$

$$\frac{\partial f_1}{\partial y} = \frac{\partial f_2}{\partial x} \quad \text{pero } \text{Dom } \bar{f} = \mathbb{R}^2 - \{(1, 0)\} \text{ no es un conjunto simplemente conexo}$$

Analizo $\oint_C \vec{f} \cdot d\vec{s}$

$$C: (x-1)^2 + y^2 = 1$$

$$\begin{cases} x-1 = \cos t \Rightarrow x = \cos t + 1 \\ y = \sin t \end{cases}$$

$$C: \vec{g}(t) = (\cos t + 1, \sin t)$$

$$t \in [0, 2\pi]$$

$$\oint_C \vec{f} \cdot d\vec{s} = \int_0^{2\pi} \left(\frac{\sin t}{\cos^2 t + \sin^2 t}, \frac{1 - \cos t - 1}{1} \right) \cdot (-\sin t, \cos t) dt$$

$$= \int_0^{2\pi} \underbrace{(-\sin^2 t - \cos^2 t)}_{-1} dt = -2\pi \neq 0$$

Luego $\vec{f}$ no admite fr. potencial (existe una curva cerrada C donde $\oint_C \vec{f} \cdot d\vec{s} \neq 0$)

$$b) \text{Dom } \bar{f} = \mathbb{R}^2 - \{0\}$$

$$\frac{\partial f_1}{\partial y} = \frac{-6(4x^2 + 9y^2) - 18y(-6y)}{(m)^2} =$$

$$= \frac{-24x^2 - 54y^2 + 108y^2}{(m)^2} = \frac{-24x^2 + 54y^2}{(m)^2}$$

$$\frac{\partial f_2}{\partial x} = \frac{6(4x^2 + 9y^2) - 8 \times 6x}{(m)^2} = \frac{24x^2 + 54y^2 - 48x^2}{(m)^2} =$$

$$= \frac{-24x^2 + 54y^2}{(m)^2}$$

$\frac{\partial f_1}{\partial y} = \frac{\partial f_2}{\partial x}$ pero el $\text{Dom } \bar{f}$ no es un conj
simplemente conexo

18) c)

$$\text{rot } \bar{f}_1 = \begin{vmatrix} \check{i} & \check{j} & \check{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{-y}{x^2+y^2} & \frac{-x}{x^2+y^2} & \varphi(z) \end{vmatrix} = i(0-0) - j(0-0) +$$

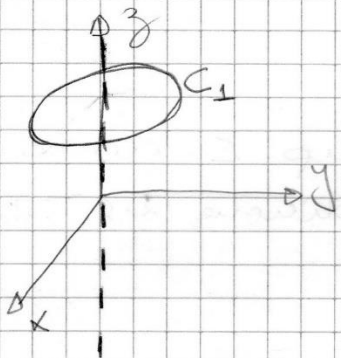
$$+ k \left(\frac{-1(x^2+y^2) - 2x(-x)}{(x^2+y^2)^2} - \frac{x^2+y^2 - 2y^2}{(x^2+y^2)^2} \right) =$$

$$= (0, 0, \frac{-x^2 - y^2 + 2x^2 - x^2 - y^2 + 2y^2}{(x^2+y^2)^2}) = (0, 0, 0) = \vec{0}$$

en $\text{Dom } \bar{f}_1$

NOTA

18)c)



$$x^2 + y^2 \neq 0 \Rightarrow x \neq 0 \vee y \neq 0$$

($\mathbb{R}^3$ menos eje $z \rightarrow$ no es simplemente conexo)

$$\text{Dom } \vec{f} = \{(x, y, z) \in \mathbb{R}^3 / x^2 + y^2 \neq 0\}$$

$$\vec{f} = \left[\frac{y}{x^2 + y^2}, \frac{-x}{x^2 + y^2}, \varphi(z) \right]$$

Sea la curva C_1 $\begin{cases} x^2 + y^2 = 1 \\ z = k \end{cases} \quad (k \in \mathbb{R})$

Si $\vec{f}$ admitiera f. potencial, debería ser

$$\oint_{C_1} \vec{f} \cdot d\vec{s} = 0. \text{ Veamos:}$$

$$C_1 \rightarrow \vec{g}(\theta) = (\cos \theta, \sin \theta, k) \quad 0 \leq \theta < 2\pi$$

$$\oint_{C_1} \vec{f} \cdot d\vec{s} = \int_0^{2\pi} \left(\frac{\sin \theta}{1}, \frac{-\cos \theta}{1}, \varphi(z) \right) (-\sin \theta, \cos \theta, 0) d\theta$$

$$= \int_0^{2\pi} (-\sin^2 \theta - \cos^2 \theta) d\theta = \int_0^{2\pi} -1 d\theta = -2\pi \neq 0$$

Luego $\vec{f}$ no admite f. potencial ya que f no es conservativo; hallé una curva cerrada C_1 tal que $\oint_{C_1} \vec{f} \cdot d\vec{s} \neq 0$

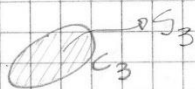
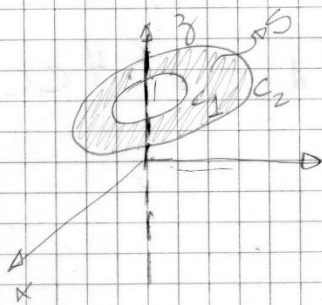
18) d)

$$\text{rot } \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{x}{x^2+y^2} & \frac{y}{x^2+y^2} & \varphi(z) \end{vmatrix} = \vec{i}(0-0) - \vec{j}(0-0) +$$

$$+ \vec{k} \left(\frac{-2xy}{(x^2+y^2)^2} - \frac{-2yx}{(x^2+y^2)^2} \right) = (0, 0, \frac{-2xy+2yx}{(x^2+y^2)^2}) =$$

$$= (0, 0, 0) = \vec{0} \quad \text{im Dom } \vec{f}.$$

$$d) \quad \vec{f} = \left[\frac{x}{x^2+y^2}, \frac{y}{x^2+y^2}, \varphi(z) \right]$$



Donde $\vec{f} = \{(x,y,z) \in \mathbb{R}^3 / x^2+y^2 \neq 0\}$
no es simpl. conexo

- Sea C_3 tal que no encierre al eje z .

$$\int_C \vec{f} \cdot d\vec{s} = \iint_{S_3} \text{rot } \vec{f} \cdot \vec{n} \, d\sigma = \iint_{S_3} \vec{0} \cdot \vec{n} \, d\sigma = 0$$

- Sea C_1 una curva cerrada que encierre al eje z , y C_2 otra curva cerrada (coplanar con C_1 que encierre al eje z y a C_1

Aplicamos el T del rotor a la sup. S .

$$\iint_S \text{rot } \vec{f} \cdot \vec{n} \, d\sigma = \int_{C_2^+} \vec{f} \cdot d\vec{s}_2 + \int_{C_1^-} \vec{f} \cdot d\vec{s}_1 = 0$$

$$\Rightarrow \iint_S \underbrace{\text{rot } \vec{f}}_{\vec{0}} \cdot \vec{n} \, d\sigma = \int_{C_2^+} \vec{f} \cdot d\vec{s}_2 - \int_{C_1^+} \vec{f} \cdot d\vec{s}_1$$

$$0 = \int_{C_2^+} \vec{f} \cdot d\vec{s}_2 - \int_{C_1^+} \vec{f} \cdot d\vec{s}_1 \Rightarrow \int_{C_1} \vec{f} \cdot d\vec{s} =$$

$$= \int_{C_2} \vec{f} \cdot d\vec{s}$$

Luego para cualquier curva C_i que "encierre" al eje z , el integral $\int_{C_i} \vec{f} \cdot d\vec{s}$ tiene el mismo valor. Calculemos una cualquiera

$$C_1 \rightarrow \vec{g}(t) = (\cos t, \sin t, 0) \quad 0 \leq t \leq 2\pi$$

$$\int_C \vec{f} \cdot d\vec{s} = \int_0^{2\pi} \left(\frac{\cos t}{1}, \frac{\sin t}{1}, \varphi(0) \right) (-\sin t, \cos t, 0) \, dt =$$

$$= \int_0^{2\pi} (\sin t \cos t + \sin t \cos t) dt = \int_0^{2\pi} 0 dt = 0$$

Por lo tanto $\forall$ curva C que encierre al eje z es

$\int_C \vec{F} d\vec{s} = 0$ y dicho círculo es nulo si no encierra al eje $z \Rightarrow \int_C \vec{F} d\vec{s} = 0 \quad \forall C \subset \mathbb{R}^3$.
 $\vec{F}$ es conservativo

19) a) $\text{rot } \vec{F} = \vec{0} \Leftrightarrow D\vec{F}$ simétrica

$$\vec{F} = (f_1, f_2, f_3)$$

$$\text{rot } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f_1 & f_2 & f_3 \end{vmatrix} = \hat{i} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) - \hat{j} \left(\frac{\partial f_3}{\partial x} - \frac{\partial f_1}{\partial z} \right) + \hat{k} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right)$$

$$\text{rot } \vec{F} = \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}, \frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x}, \frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right)$$

$$\text{rot } \vec{F} = \vec{0} \Rightarrow \begin{cases} \frac{\partial f_3}{\partial y} = \frac{\partial f_2}{\partial z} & (1) \\ \frac{\partial f_1}{\partial z} = \frac{\partial f_3}{\partial x} & (2) \\ \frac{\partial f_2}{\partial x} = \frac{\partial f_1}{\partial y} & (3) \end{cases}$$

$\Uparrow \Downarrow$

$$D\vec{F} = \begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial y} & \frac{\partial f_3}{\partial z} \end{pmatrix}$$

$\bigcirc$ iguales por (3)

$\square$ iguales por (2)

$\triangle$ iguales por (1)

$\Downarrow \Uparrow$

$$D\vec{F} = (D\vec{F})^t = {}^t D\vec{F} \text{ simétrica}$$

$$b) \vec{f} \in C^2 \Rightarrow \operatorname{div}(\operatorname{rot} \vec{f}) \equiv 0$$

$$\operatorname{rot} \vec{f} = \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}, \frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x}, \frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right)$$

$$\operatorname{div} \operatorname{rot} \vec{f} = \frac{\partial}{\partial x} \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) =$$

$$= \frac{\partial^2 f_3}{\partial x \partial y} - \frac{\partial^2 f_2}{\partial x \partial z} + \frac{\partial^2 f_1}{\partial y \partial z} - \frac{\partial^2 f_3}{\partial y \partial x} + \frac{\partial^2 f_2}{\partial z \partial x} - \frac{\partial^2 f_1}{\partial z \partial y} = 0$$

$$\downarrow \quad \quad \quad \uparrow \quad \quad \quad \uparrow$$

$$= \operatorname{rot} \vec{f} \in C^2 \text{ y T. Schwarz} = \operatorname{rot} \vec{f} \in C^2 \text{ y T. Schwarz}$$

$$= \operatorname{rot} \vec{f} \in C^2 \text{ y T. Schwarz}$$

c) $\bar{f} = \nabla \phi \in C^1 \Rightarrow \bar{f} = (\phi'_x, \phi'_y, \phi'_z)$

$\text{rot } \bar{f} = \begin{vmatrix} \overset{i}{\frac{\partial}{\partial x}} & \overset{j}{\frac{\partial}{\partial y}} & \overset{k}{\frac{\partial}{\partial z}} \\ \phi'_x & \phi'_y & \phi'_z \end{vmatrix} = \overset{0}{i} (\phi''_{zy} - \phi''_{yz}) - \overset{0}{j} (\phi''_{zx} - \phi''_{xz}) + \overset{0}{k} (\phi''_{yx} - \phi''_{xy}) = (0, 0, 0)$

$\phi''_{ij} = \phi''_{ji}$ pot. Schwarz $\text{rot } \bar{f} = \bar{0}$

d) $\text{div}(a\bar{f} + b\bar{g}) = \frac{\partial}{\partial x}(af_1 + bg_1) + \frac{\partial}{\partial y}(af_2 + bg_2) + \frac{\partial}{\partial z}(af_3 + bg_3) = a \frac{\partial f_1}{\partial x} + b \frac{\partial g_1}{\partial x} + a \frac{\partial f_2}{\partial y} + b \frac{\partial g_2}{\partial y} + a \frac{\partial f_3}{\partial z} + b \frac{\partial g_3}{\partial z} = a \left(\frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} \right) + b \left(\frac{\partial g_1}{\partial x} + \frac{\partial g_2}{\partial y} + \frac{\partial g_3}{\partial z} \right) = a \text{div } \bar{f} + b \text{div } \bar{g}$

$\text{Rot}(a\bar{f} + b\bar{g}) = \begin{vmatrix} \overset{i}{\frac{\partial}{\partial x}} & \overset{j}{\frac{\partial}{\partial y}} & \overset{k}{\frac{\partial}{\partial z}} \\ af_1 + bg_1 & af_2 + bg_2 & af_3 + bg_3 \end{vmatrix} = \overset{i}{\left(a \frac{\partial f_3}{\partial y} + b \frac{\partial g_3}{\partial y} - a \frac{\partial f_2}{\partial z} - b \frac{\partial g_2}{\partial z} \right)} - \overset{j}{\left(a \frac{\partial f_3}{\partial x} + b \frac{\partial g_3}{\partial x} - a \frac{\partial f_1}{\partial z} - b \frac{\partial g_1}{\partial z} \right)} + \overset{k}{\left(a \frac{\partial f_2}{\partial x} + b \frac{\partial g_2}{\partial x} - a \frac{\partial f_1}{\partial y} - b \frac{\partial g_1}{\partial y} \right)}$

$= a \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z}, \frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x}, \frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) + b \left(\frac{\partial g_3}{\partial y} - \frac{\partial g_2}{\partial z}, \frac{\partial g_1}{\partial z} - \frac{\partial g_3}{\partial x}, \frac{\partial g_2}{\partial x} - \frac{\partial g_1}{\partial y} \right) = a \text{rot } \bar{f} + b \text{rot } \bar{g}$

$$\begin{aligned}
 e) \operatorname{div}(f \cdot \vec{g}) &= \operatorname{div}(fg_1, fg_2, fg_3) = \\
 &= \frac{\partial(fg_1)}{\partial x} + \frac{\partial(fg_2)}{\partial y} + \frac{\partial(fg_3)}{\partial z} = \underbrace{f'_x g_1 + f \frac{\partial g_1}{\partial x}} + \underbrace{f'_y g_2 + f \frac{\partial g_2}{\partial y}} + \underbrace{f'_z g_3 + f \frac{\partial g_3}{\partial z}} = \nabla f \cdot (g_1, g_2, g_3) + \\
 &+ f \left(\frac{\partial g_1}{\partial x} + \frac{\partial g_2}{\partial y} + \frac{\partial g_3}{\partial z} \right) = \nabla f \cdot \vec{g} + f \operatorname{div} \vec{g}
 \end{aligned}$$

$$\begin{aligned}
 \operatorname{rot}(f \cdot \vec{g}) &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ fg_1 & fg_2 & fg_3 \end{vmatrix} = \vec{i} \left(\frac{\partial(fg_3)}{\partial y} - \frac{\partial(fg_2)}{\partial z} \right) - \\
 &- \vec{j} \left(\frac{\partial(fg_3)}{\partial x} - \frac{\partial(fg_1)}{\partial z} \right) + \vec{k} \left(\frac{\partial(fg_2)}{\partial x} - \frac{\partial(fg_1)}{\partial y} \right) = \\
 &= (f'_y g_3 + f \frac{\partial g_3}{\partial y} - f'_z g_2 - f \frac{\partial g_2}{\partial z}, f'_z g_1 + f \frac{\partial g_1}{\partial z} - f'_x g_3 - f \frac{\partial g_3}{\partial x}, f'_x g_2 + f \frac{\partial g_2}{\partial x} - f'_y g_1 - f \frac{\partial g_1}{\partial y}) = \\
 &= f \left(\frac{\partial g_3}{\partial y} - \frac{\partial g_2}{\partial z}, \frac{\partial g_1}{\partial z} - \frac{\partial g_3}{\partial x}, \frac{\partial g_2}{\partial x} - \frac{\partial g_1}{\partial y} \right) + \\
 &+ (f'_y g_3 - f'_z g_2, f'_z g_1 - f'_x g_3, f'_x g_2 - f'_y g_1) = \\
 &= f \cdot \operatorname{rot} \vec{g} + \nabla f \times \vec{g}
 \end{aligned}$$

$$\begin{aligned}
 f) \operatorname{div}(\vec{f} \wedge \vec{g}) &= \operatorname{div} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ f_1 & f_2 & f_3 \\ g_1 & g_2 & g_3 \end{vmatrix} = i(f_2 g_3 - f_3 g_2) - \\
 &- j(f_1 g_3 - f_3 g_1) + k(f_1 g_2 - f_2 g_1) = \\
 &= \frac{\partial}{\partial x}(f_2 g_3 - f_3 g_2) + \frac{\partial}{\partial y}(f_3 g_1 - f_1 g_3) + \frac{\partial}{\partial z}(f_1 g_2 - f_2 g_1) = \\
 &= \frac{\partial f_2}{\partial x} g_3 + f_2 \frac{\partial g_3}{\partial x} - \frac{\partial f_3}{\partial x} g_2 - f_3 \frac{\partial g_2}{\partial x} + \frac{\partial f_3}{\partial y} g_1 + f_3 \frac{\partial g_1}{\partial y} - \\
 &- \frac{\partial f_1}{\partial y} g_3 - f_1 \frac{\partial g_3}{\partial y} + \frac{\partial f_1}{\partial z} g_2 + f_1 \frac{\partial g_2}{\partial z} - \frac{\partial f_2}{\partial z} g_1 - f_2 \frac{\partial g_1}{\partial z} =
 \end{aligned}$$

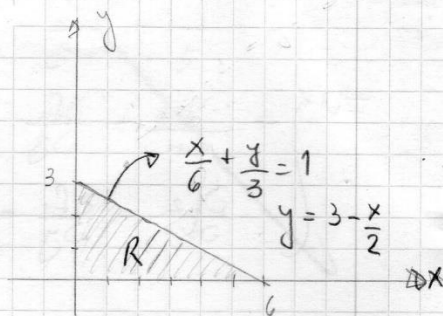
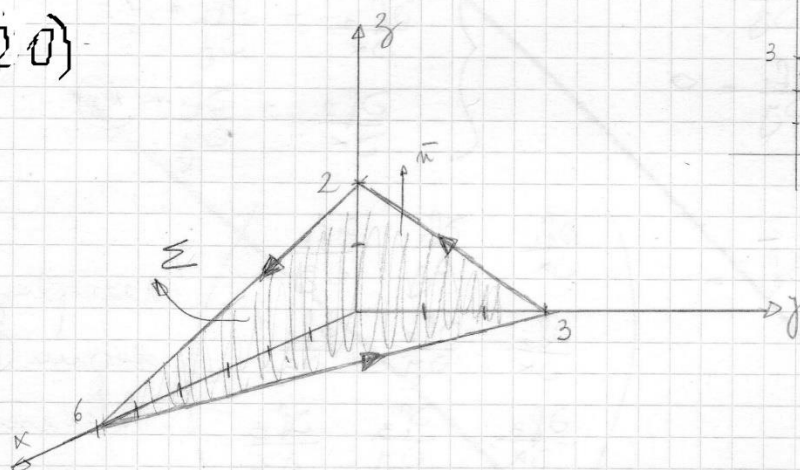
$$\vec{g} \cdot (-\operatorname{rot} \vec{f})$$

$$\begin{aligned}
 &= g_1 \left(\frac{\partial f_3}{\partial y} - \frac{\partial f_2}{\partial z} \right) + g_2 \left(\frac{\partial f_1}{\partial z} - \frac{\partial f_3}{\partial x} \right) + g_3 \left(\frac{\partial f_2}{\partial x} - \frac{\partial f_1}{\partial y} \right) + \\
 &+ f_1 \left(\frac{\partial g_2}{\partial z} - \frac{\partial g_3}{\partial y} \right) + f_2 \left(\frac{\partial g_3}{\partial x} - \frac{\partial g_1}{\partial z} \right) + f_3 \left(\frac{\partial g_1}{\partial y} - \frac{\partial g_2}{\partial x} \right) =
 \end{aligned}$$

$$\vec{f} \cdot \operatorname{rot} \vec{g}$$

$$= \vec{f} \cdot \operatorname{rot} \vec{g} - \vec{g} \cdot \operatorname{rot} \vec{f}$$

20)



$$x + 2y + 3z = 6 \Rightarrow \frac{x}{6} + \frac{y}{3} + \frac{z}{2} = 1$$

$$\oint_C \vec{f} \cdot d\vec{s} = \iint_{\Sigma} \text{rot } \vec{f} \cdot \vec{n} \, d\sigma = \textcircled{A}$$

$$\text{rot } \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x-y & x+y & 3-x-y \end{vmatrix} = \vec{i}(-1) - \vec{j}(-1) + \vec{k}(1+1) = (-1, 1, 2)$$

$$G(x, y, z) = x + 2y + 3z - 6 = 0$$

$$\vec{n} = \frac{\nabla G}{\|\nabla G\|} = \frac{(1, 2, 3)}{\sqrt{1+4+9}} = \frac{(1, 2, 3)}{\sqrt{14}} \quad (\text{exterior})$$

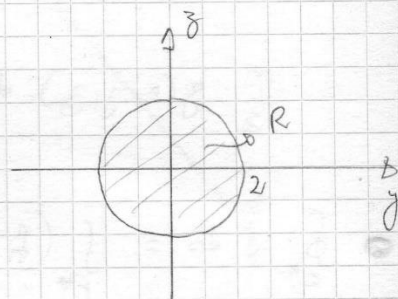
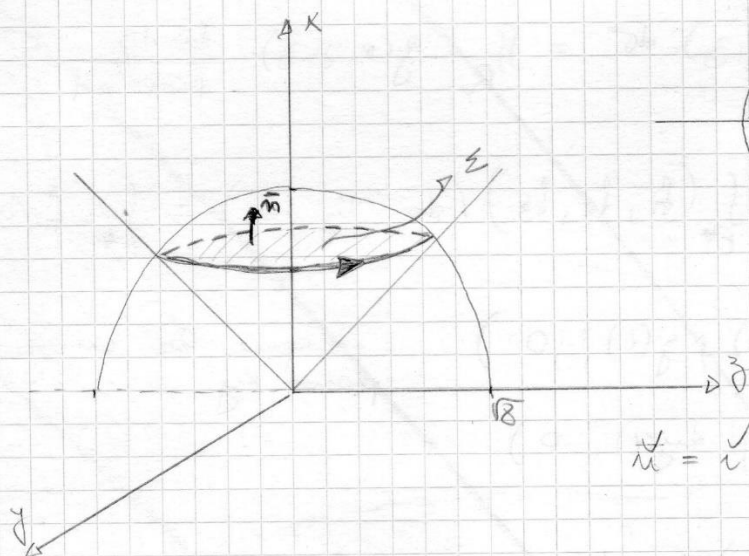
$$⑦ = \iint_{\Sigma} (-1, 1, 2) \cdot \frac{(1, 2, 3)}{\sqrt{14}} d\sigma = \iint_{\Sigma} \frac{-1+2+6}{\sqrt{14}} d\sigma =$$

$$= \frac{7}{\sqrt{14}} \iint_{\Sigma} d\sigma = \frac{7}{\sqrt{14}} \iint_R \frac{\sqrt{1+4+9}}{|3|} dx dy = \frac{7}{\sqrt{14}} \iint_R \frac{\sqrt{14}}{3} dx dy =$$

$$= \frac{7}{3} \int_0^6 dx \int_0^{3-\frac{x}{2}} dy = \frac{7}{3} \int_0^6 \left(3 - \frac{x}{2}\right) dx =$$

$$= \frac{7}{3} \left[3x - \frac{x^2}{4} \right]_0^6 = \frac{7}{3} \left[18 - \frac{36}{4} \right] = \frac{7}{3} \cdot \frac{63}{4} = \frac{21}{4}$$

21)



$$\vec{n} = \vec{v}$$

$$x^2 + y^2 + z^2 = 8 \Rightarrow y^2 + z^2 + y^2 + z^2 = 8 \Rightarrow z^2 + y^2 = 4$$

$$x = \sqrt{y^2 + z^2}$$

$$\oint_C \vec{f} \cdot d\vec{s} = \iint_{\Sigma} \text{rot } \vec{f} \cdot \vec{n} \, d\sigma = \iint_{\Sigma} (z^2, 0, -1-x) \cdot (4, 0, 0) \, d\sigma =$$

$$\text{rot } \vec{f} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & y-x & z^2 \end{vmatrix} = \vec{i}(z^2) - \vec{j}(0) + \vec{k}(-1-x) = (z^2, 0, -1-x)$$

le sup ne se déforme en la proyección

$$= \iint_{\Sigma} z^2 \, d\sigma = \iint_R z^2 \cdot 1 \, dy \, dz = \iint_R z^2 \, dy \, dz =$$

$$= \int_0^{2\pi} d\theta \int_0^2 \rho^2 \sin^2 \theta \, \rho \, d\rho = \int_0^{2\pi} \sin^2 \theta \, d\theta \left[\frac{\rho^4}{4} \right]_0^2 =$$

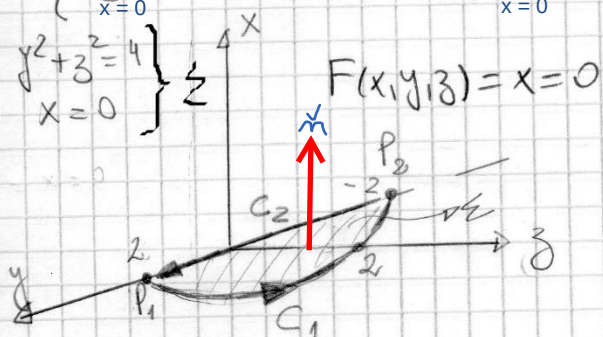
$$= \int_0^{2\pi} \sin^2 \theta \, 4 \, d\theta = 4 \left[\frac{1}{2} \theta - \frac{1}{4} \sin 2\theta \right]_0^{2\pi} =$$

$$= 4 \left[\frac{1}{2} \cdot 2\pi - 0 - 0 + 0 \right] = 4\pi$$

Siendo $\vec{f}$ de clase C^1 , $\text{rot}(\vec{f}(x,y,z)) = (3, 1, 2y)$, calcule la circulación de $\vec{f}$ a lo largo del arco de curva de ecuación $\vec{X} = (0, 2\cos u, 2\sin u)$ con $u \in [0, \pi]$, sabiendo que la circulación de $\vec{f}$ por el segmento desde $(0, -2, 0)$ hasta $(0, 2, 0)$ es igual a $16/3$

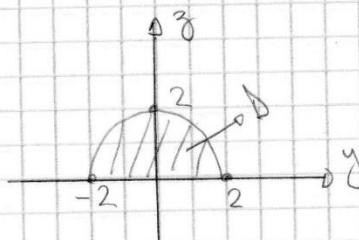
$$22) \quad \vec{X} = (0, 2\cos u, 2\sin u)$$

$$\begin{cases} y = 2\cos u \\ z = 2\sin u \end{cases} \Rightarrow \begin{cases} y^2 = 4\cos^2 u \\ z^2 = 4\sin^2 u \end{cases} \Rightarrow y^2 + z^2 = 4 \quad \text{con } x=0$$



$$u=0 \Rightarrow P_1 = (0, 2, 0)$$

$$u=\pi \Rightarrow P_2 = (0, -2, 0)$$



$$C = C_1 \cup C_2$$

$$\oint_C \vec{f} \cdot d\vec{s} = \iint_{\Sigma} \text{rot} \vec{f} \cdot \vec{n} \, d\sigma$$

$$\int_{C_1} \vec{f} \cdot d\vec{s} + \int_{C_2} \vec{f} \cdot d\vec{s} = \iint_{\Sigma} (3, 1, 2y) \cdot (1, 0, 0) \, d\sigma = \iint_{\Sigma} 3 \, d\sigma =$$

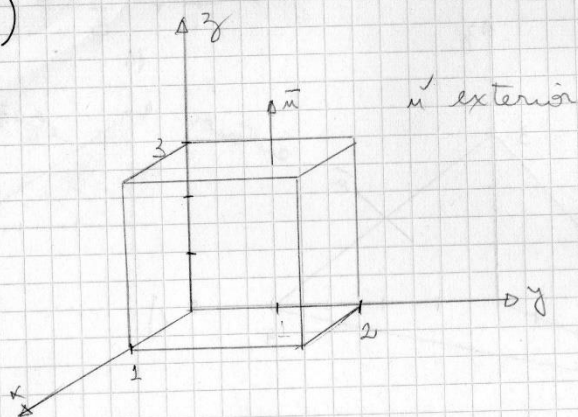
$$= 3 \iint_D \frac{\sqrt{F_x^2 + F_y^2 + F_z^2}}{|F_x|} \, dz \, dy = 3 \iint_D \frac{\sqrt{1}}{1} \, dz \, dy =$$

$$= 3 \int_0^{\pi} d\theta \int_0^2 \rho \, \rho \, d\rho = 3\pi \left[\frac{\rho^2}{2} \right]_0^2 = 3\pi \cdot \frac{4}{2} = 6\pi$$

$$\int_{C_1} \vec{f} \cdot d\vec{s} + \int_{C_2} \vec{f} \cdot d\vec{s} = 6\pi \Rightarrow \int_{C_1} \vec{f} \cdot d\vec{s} = 6\pi - \underbrace{\int_{C_2} \vec{f} \cdot d\vec{s}}_{\frac{16}{3}} =$$

$$\Rightarrow \int_{C_1} \vec{f} \cdot d\vec{s} = 6\pi - \frac{16}{3}$$

23)



$$\begin{aligned}
 \iint_{\Sigma} \vec{F} \cdot \vec{n} \, d\sigma &= \iiint_V \operatorname{div} \vec{F} \, dx \, dy \, dz = \iiint_V (y + z + x) \, dx \, dy \, dz = \\
 &= \int_0^1 dx \int_0^2 dy \int_0^3 dz (x + y + z) = \int_0^1 dx \int_0^2 dy \left[xz + yz + \frac{z^2}{2} \Big|_0^3 \right] = \\
 &= \int_0^1 dx \int_0^2 dy \left[3x + 3y + \frac{9}{2} \right] = \int_0^1 dx \left[3xy + \frac{3y^2}{2} + \frac{9}{2}y \Big|_0^2 \right] = \\
 &= \int_0^1 \left[6x + \frac{3 \cdot 4}{2} + \frac{9}{2} \cdot 2 \right] dx = \left[6 \frac{x^2}{2} + 15x \Big|_0^1 \right] = 3 + 15 = 18
 \end{aligned}$$

$$\iint_{\Sigma} \vec{F} \cdot \vec{n} \, d\sigma = \iint_{\Sigma_1} + \iint_{\Sigma_2} + \iint_{\Sigma_3} + \iint_{\Sigma_4} + \iint_{\Sigma_5} + \iint_{\Sigma_6} \quad \text{--- (A)}$$

$$\bullet \Sigma_1 \rightarrow \begin{cases} 0 \leq x \leq 1 \\ 0 \leq y \leq 2 \\ z = 0 \end{cases} \quad \vec{n} = -\vec{k}$$

$$\begin{aligned}
 \iint_{\Sigma_1} \vec{F} \cdot \vec{n} \, d\sigma_1 &= \iint_{\Sigma_1} (x, y, z) \cdot (0, 0, -1) \, d\sigma_1 = \iint_{\Sigma_1} -x \, d\sigma_1 = \\
 &= \iint_{\Sigma_1} -x \cdot 0 \, d\sigma_1 = 0
 \end{aligned}$$

$$\bullet \Sigma_2 \rightarrow \begin{cases} 0 \leq x \leq 1 \\ 0 \leq y \leq 2 \\ z = 3 \end{cases} \quad \vec{n} = \vec{k}$$

$$\begin{aligned}
 \iint_{\Sigma_2} \vec{F} \cdot \vec{n} \, d\sigma_2 &= \iint_{\Sigma_2} (x, y, z) \cdot (0, 0, 1) \, d\sigma_2 = \iint_{\Sigma_2} x \, d\sigma_2 = \\
 &= \iint_{\Sigma_2} x \, d\sigma_2 = 3 \int_0^1 dx \int_0^2 dy x = 3 \int_0^1 2x \, dx = 3 \left[x^2 \Big|_0^1 \right] = 3
 \end{aligned}$$

$$\bullet \Sigma_3 \rightarrow \begin{cases} 0 \leq x \leq 1 \\ 0 \leq z \leq 3 \\ y=0 \end{cases} \quad \vec{n} = -\vec{j}$$

$$\iint_{\Sigma_3} \vec{F} \cdot \vec{n} \, d\sigma_3 = \iint_{\Sigma_3} (x y, y z, x z) (0, -1, 0) \, d\sigma_3 = \iint_{\Sigma_3} -y z \, d\sigma_3 =$$

$$= \iint_{\Sigma_3} -0 \cdot z \, d\sigma_3 = 0$$

$$\bullet \Sigma_4 \rightarrow \begin{cases} 0 \leq x \leq 1 \\ 0 \leq z \leq 3 \\ y=2 \end{cases} \quad \vec{n} = \vec{j}$$

$$\iint_{\Sigma_4} \vec{F} \cdot \vec{n} \, d\sigma_4 = \iint_{\Sigma_4} (x y, y z, x z) (0, 1, 0) \, d\sigma_4 = \iint_{\Sigma_4} y z \, d\sigma_4 =$$

$$= \iint_{\Sigma_4} 2 z \, d\sigma_4 = 2 \int_0^1 dx \int_0^3 dz z = 2 \cdot 1 \cdot \left[\frac{z^2}{2} \Big|_0^3 \right] = 9$$

$$\bullet \Sigma_5 \rightarrow \begin{cases} 0 \leq y \leq 2 \\ 0 \leq z \leq 3 \\ x=0 \end{cases} \quad \vec{n} = -\vec{i}$$

$$\iint_{\Sigma_5} \vec{F} \cdot \vec{n} \, d\sigma_5 = \iint_{\Sigma_5} (x y, y z, x z) (-1, 0, 0) \, d\sigma_5 = \iint_{\Sigma_5} -x y \, d\sigma_5 =$$

$$= \iint_{\Sigma_5} -0 \cdot y \, d\sigma_5 = 0$$

$$\bullet \Sigma_6 \rightarrow \begin{cases} 0 \leq y \leq 2 \\ 0 \leq z \leq 3 \\ x=1 \end{cases} \quad \vec{n} = \vec{i}$$

$$\iint_{\Sigma_6} \vec{F} \cdot \vec{n} \, d\sigma_6 = \iint_{\Sigma_6} (x y, y z, x z) (1, 0, 0) \, d\sigma_6 = \iint_{\Sigma_6} x y \, d\sigma_6 =$$

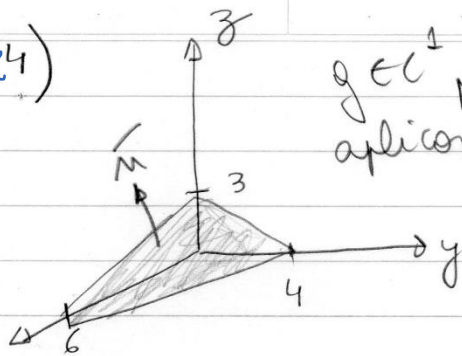
$$= \iint_{\Sigma_6} y \, d\sigma_6 = \int_0^2 dy \int_0^3 dz y = \int_0^2 3 y \, dy = \left[\frac{3y^2}{2} \Big|_0^2 \right] = 6$$

Reemplazando los resultados obtenidos en (A) :

$$\iint_{\Sigma} \vec{F} \cdot \vec{n} \, d\sigma = 0 + 3 + 0 + 9 + 0 + 6 = 18$$

Calcule el flujo de $\vec{f}(x,y,z)=(x-y-z, y-x-z, g(x,y))$ a través de la superficie frontera del cuerpo definido por $2x+3y+4z \leq 12$ en el 1er. octante. Indique la orientación del vector normal que considera y las hipótesis que supone para el campo escalar g

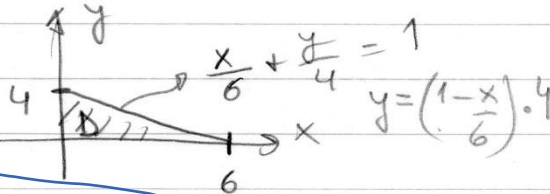
24)



$g \in C^1$ para
aplicar T. dir.

$$2x+3y+4z=12$$

$$\frac{x}{6} + \frac{y}{4} + \frac{z}{3} = 1$$



$$\operatorname{div} \vec{f} = \nabla \cdot \vec{f} = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right) \cdot (x-y-z, y-x-z, g(x,y))$$

$$\int_S \vec{f} \cdot \vec{n} \, dS = \iiint_V \operatorname{div} \vec{f} \, dx \, dy \, dz = \iiint_V (1+1+0) \, dx \, dy \, dz =$$

$$= 2 \int_0^6 dx \int_0^{4-\frac{2}{3}x} dy \int_0^{\frac{12-2x-3y}{4}} dz =$$

$$= 2 \int_0^6 dx \int_0^{4-\frac{2}{3}x} dy \left(3 - \frac{x}{2} - \frac{3}{4}y \right) =$$

$$= 2 \int_0^6 dx \left[3y - \frac{x}{2}y - \frac{3}{8}y^2 \right]_0^{4-\frac{2}{3}x} =$$

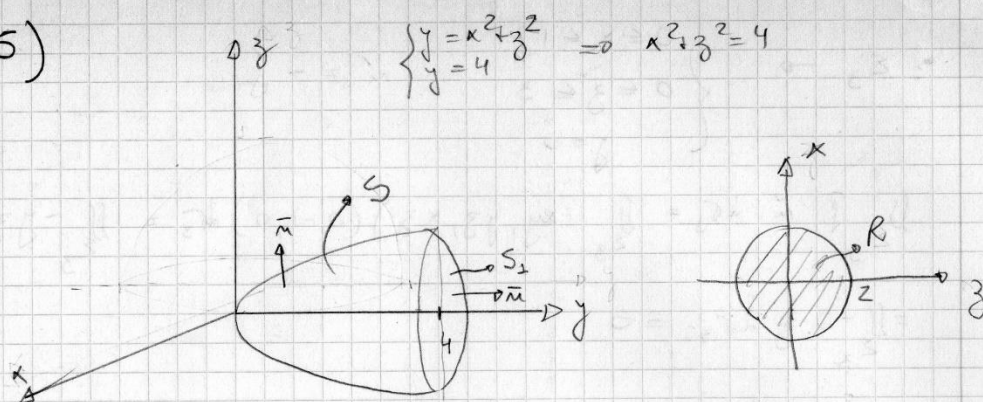
$$= 2 \int_0^6 dx \left[3\left(4-\frac{2}{3}x\right) - \frac{x}{2}\left(4-\frac{2}{3}x\right) - \frac{3}{8}\left(4-\frac{2}{3}x\right)^2 \right] =$$

$$= 2 \int_0^6 \left[\cancel{12} - \cancel{2x} - 2x + \frac{x^2}{3} - \cancel{6} - \frac{1}{6}x^2 + \cancel{2x} \right] dx$$

$$= 2 \int_0^6 \left(6 - 2x + \frac{1}{6}x^2 \right) dx = 2 \left[6x - x^2 + \frac{1}{18}x^3 \right]_0^6 =$$

$$= 2 \left(\cancel{36} - \cancel{36} + \frac{12^3}{18} \right) = (24)$$

225)



$$S_2 = S \cup S_1 \quad S_2 \rightarrow \text{sup. cerrada}$$

$$\iint_S \vec{F} \cdot \vec{n}_1 d\vec{s}_1 + \iint_{S_1} \vec{F} \cdot \vec{n}_1 d\vec{s}_1 = \iint_{S_2} \vec{F} \cdot \vec{n}_2 d\vec{s}_2$$

$$\iint_{S_2} \vec{F} \cdot \vec{n}_2 d\vec{s}_2 = \iiint_V \text{div} \vec{F} \, dx \, dy \, dz =$$

$$= \iiint_V (2xz^2 + xz^2 - 3xz^2) \, dx \, dy \, dz = \iiint_V 0 \, dx \, dy \, dz = 0$$

$$\text{Luego: } \iint_S \vec{F} \cdot \vec{n} \, d\vec{s} = - \iint_{S_1} \vec{F} \cdot \vec{n}_1 \, d\vec{s}_1 = - \iint_R (x^2 z^2, 1 + xy z^2, 1 - x z^3) \cdot$$

$$(0, 1, 0) \, dx \, dz = - \iint_R (1 + xy z^2) \, dx \, dz =$$

la proyección no se deforma

$$= - \iint_R (1 + 4xz^2) \, dx \, dz = - \int_0^{2\pi} d\theta \int_0^2 \rho \, \rho (1 + 4\rho \sin\theta \rho^2 \cos^2\theta) \, \rho \, d\rho =$$

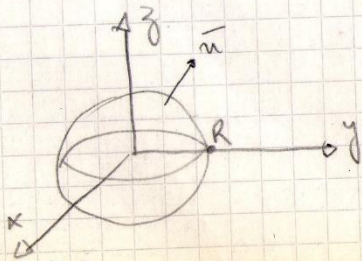
$$= - \int_0^{2\pi} d\theta \int_0^2 (\rho + 4\rho^4 \sin\theta \cos^2\theta) \, \rho \, d\rho = - \int_0^{2\pi} d\theta \left[\frac{\rho^2}{2} + \frac{4\rho^5}{5} \sin\theta \cos^2\theta \right]_0^2 =$$

$$= - \int_0^{2\pi} d\theta \left[\frac{4}{2} + \frac{4 \cdot 32}{5} \sin\theta \cos^2\theta \right] =$$

$$= - \left[2\theta - \frac{128}{5} \frac{\cos^3\theta}{3} \right]_0^{2\pi} = - \left[2 \cdot 2\pi - \frac{128}{5} \frac{1}{3} - 0 + \frac{128}{5} \cdot \frac{1}{3} \right] =$$

$$= -4\pi$$

26)



$$\iint_S \vec{F} \cdot \vec{n} \, d\sigma = \iiint_V \operatorname{div} \vec{F} \, dx \, dy \, dz = \iiint_V (\cancel{4(x \cdot z)} + \cancel{x})$$

$$\cdot (\cancel{4(x \cdot z)} + 2y - \cancel{4(x \cdot z)} - z \cdot (\cancel{4(x \cdot z)} \times)) \, dx \, dy \, dz = \iiint_V 2y \, dx \, dy \, dz =$$

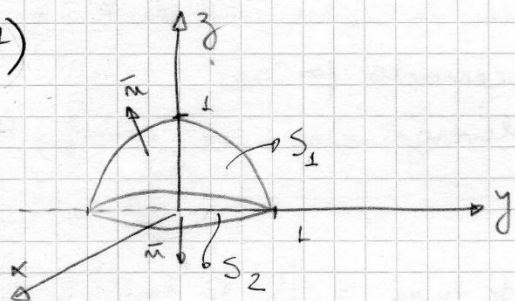
$$= \int_0^{2\pi} d\theta \int_0^R \rho \, d\rho \int_0^\pi d\lambda \, \rho^2 \sin \lambda \, 2 \rho \sin \theta \sin \lambda =$$

$$= \int_0^{2\pi} d\theta \sin \theta \int_0^R \rho \, d\rho \int_0^\pi d\lambda \sin^2 \lambda = \left[-\cos \theta \right]_0^{2\pi} \cdot$$

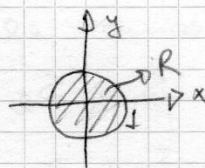
$$\cdot \left[\frac{\rho^4}{4} \right]_0^R \left[\frac{1}{2} \lambda - \frac{1}{4} \sin 2\lambda \right]_0^\pi = [-1 + 1] \cdot \frac{R^4}{4} \cdot$$

$$\cdot \left[\frac{1}{2} \pi - 0 \right] = 0$$

27)



$$S = S_1 \cup S_2$$



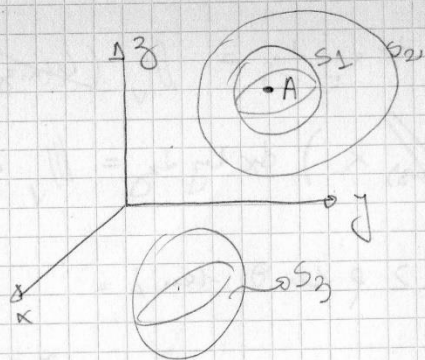
$$\iint_S \vec{F} \cdot \vec{n} \, d\sigma = \iint_{S_1} \vec{F} \cdot \vec{n}_1 \, d\sigma_1 + \iint_{S_2} \vec{F} \cdot \vec{n}_2 \, d\sigma_2$$

en la proyección
el vector
no se deforma

$$\iint_V \text{div } \vec{F} \, dx \, dy \, dz = \iint_{S_1} \vec{F} \cdot \vec{n}_1 \, d\sigma_1 + \iint_R (x, y, x^2) (0, 0, -1) \, dx \, dy$$

$$\begin{aligned} \Rightarrow \iint_{S_1} \vec{F} \cdot \vec{n}_1 \, d\sigma_1 &= \iint_V 2(1+z) \, dx \, dy \, dz - \iint_R -x^2 \, dx \, dy = \\ &= 2 \int_0^{2\pi} d\theta \int_0^{\pi/2} d\lambda \int_0^1 \rho^2 \sin \lambda (1 + \rho \cos \lambda) + \\ &+ \int_0^{2\pi} d\theta \int_0^1 \rho^2 \cos^2 \theta \, d\rho = \\ &= 2 \cdot 2\pi \int_0^{\pi/2} d\lambda \sin \lambda \left[\frac{\rho^3}{3} + \frac{\rho^4}{4} \cos \lambda \right]_0^1 + \\ &+ \int_0^{2\pi} \cos^2 \theta \, d\theta \left[\frac{\rho^4}{4} \right]_0^1 = \\ &= 4\pi \int_0^{\pi/2} d\lambda \sin \lambda \left[\frac{1}{3} + \frac{\cos \lambda}{4} \right] + \frac{1}{4} \left[\frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right]_0^{2\pi} = \\ &= 4\pi \left[-\frac{\cos \lambda}{3} + \frac{\sin^2 \lambda}{2 \cdot 4} \right]_0^{\pi/2} + \frac{1}{4} \left[\frac{1}{2} \cdot 2\pi + 0 - 0 \right] = \\ &= 4\pi \left[0 + \frac{1}{8} + \frac{1}{3} - 0 \right] + \frac{\pi}{4} = 4 \cdot \frac{11\pi}{24} + \frac{\pi}{4} = \frac{25\pi}{12} \end{aligned}$$

28)



- Sea S_3 una sup cerrada que no contiene al pto A

$$\iint_{S_3} \vec{F} \cdot \vec{n} \, d\sigma = \iiint_V \operatorname{div} \vec{F} \, dx \, dy \, dz = \iiint_V 0 \, dx \, dy \, dz = 0$$

- Sea S_2 una sup cerrada que contiene al pto A , y sea una sup auxiliar S_1 que encierra al pto A y esto incluido en el vol V encerrado por S_2

Aplicamos el T. de la divergencia al cuerpo H ubicado entre S_1 y S_2

$$\iint_{\partial H} \vec{F} \cdot \vec{n} \, d\sigma = \iiint_H \operatorname{div} \vec{F} \, dx \, dy \, dz = \iiint_H 0 \, dx \, dy \, dz = 0$$

$$\iint_{\partial H} \vec{F} \cdot \vec{n} \, d\sigma = 0 \Rightarrow \iint_{S_2} \vec{F} \cdot \vec{n} \, d\sigma - \iint_{S_1} \vec{F} \cdot \vec{n} \, d\sigma = 0 \Rightarrow$$

$$\Rightarrow \iint_{S_2} \vec{F} \cdot \vec{n} \, d\sigma = \iint_{S_1} \vec{F} \cdot \vec{n} \, d\sigma \quad \text{Sea } k = \iint_{S_1} \vec{F} \cdot \vec{n} \, d\sigma = \iint_{S_2} \vec{F} \cdot \vec{n} \, d\sigma$$

Luego : i) Si S no encierra al pto $A \Rightarrow \iint_S \vec{F} \cdot \vec{n} \, d\sigma = 0$

ii) Si S encierra al pto $A \Rightarrow \iint_S \vec{F} \cdot \vec{n} \, d\sigma = k$ para

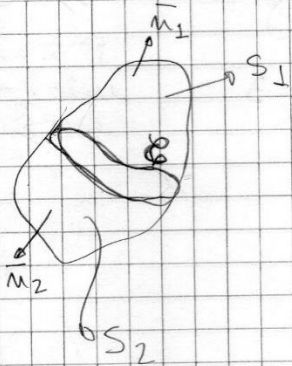
cualquiera que sea la sup S

30) $\vec{f} = \text{rot } \vec{g}$ $\vec{g}$ es potencial vectorial de $\vec{f}$

$$\vec{f}: \mathbb{R}^3 \rightarrow \mathbb{R}^3 ; \vec{g}: \mathbb{R}^3 \rightarrow \mathbb{R}^3 ; \vec{f} \in C^1$$

S sup. cerrada

$$\iint_S \vec{f} \cdot \vec{n} \, d\sigma = \iint_S \text{rot } \vec{g} \cdot \vec{n} \, d\sigma = \textcircled{A}$$



Si S es cerrada, puedo dividirlo en S_1 y S_2 , limitados por la curva C . Elijo $\vec{n}$ conveniente para el cálculo

$$\textcircled{A} = \iint_{S_1} \text{rot } \vec{g} \cdot \vec{n}_1 \, d\sigma_1 + \iint_{S_2} \text{rot } \vec{g} \cdot \vec{n}_2 \, d\sigma_2 =$$

$$= \oint_{C^+} \vec{g} \cdot d\vec{s} + \oint_{C^-} \vec{g} \cdot d\vec{s} = \oint_{C^+} \vec{g} \cdot d\vec{s} - \oint_{C^+} \vec{g} \cdot d\vec{s} =$$

× teorema del rotor

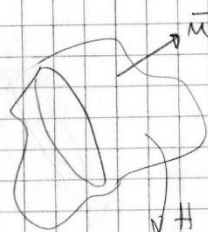
$$= 0$$

31) ϕ armónico $\Rightarrow \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = 0$
 $\operatorname{div}(\nabla \phi) = 0$

$$\vec{J} = \phi \cdot \nabla \phi$$

$$\iint_{\partial H} \vec{J} \cdot \vec{n} \, d\sigma = \iint_{\partial H} \phi \cdot \nabla \phi \cdot \vec{n} \, d\sigma =$$

$$= \iiint_H \operatorname{div}(\phi \cdot \nabla \phi) \, dx \, dy \, dz =$$



∂H (H es cerrado
que es un
cuerpo)

valor
diverg

$$= \iiint_H \operatorname{div}(\phi \phi_x, \phi \phi_y, \phi \phi_z) \, dx \, dy \, dz =$$

$$= \iiint_H (\phi_x' \phi_x' + \phi (\phi''_{xx}) + \phi_y' \phi_y' + \phi (\phi''_{yy}) + \phi_z' \phi_z' + \phi (\phi''_{zz})) \, dx \, dy \, dz =$$

$$= \iiint_H \left[(\phi_x'^2 + \phi_y'^2 + \phi_z'^2) + \phi (\phi''_{xx} + \phi''_{yy} + \phi''_{zz}) \right] \, dx \, dy \, dz =$$

$= 0$ por armónico

$$= \iiint_H \left[(\phi_x'^2 + \phi_y'^2 + \phi_z'^2) + \phi \cdot 0 \right] \, dx \, dy \, dz =$$

$$= \iiint_H (\phi_x'^2 + \phi_y'^2 + \phi_z'^2) \, dx \, dy \, dz$$

$\geq 0 \quad \forall (x, y, z) \in \mathbb{R}^3$

≥ 0
 porque el
 integrando
 es ≥ 0

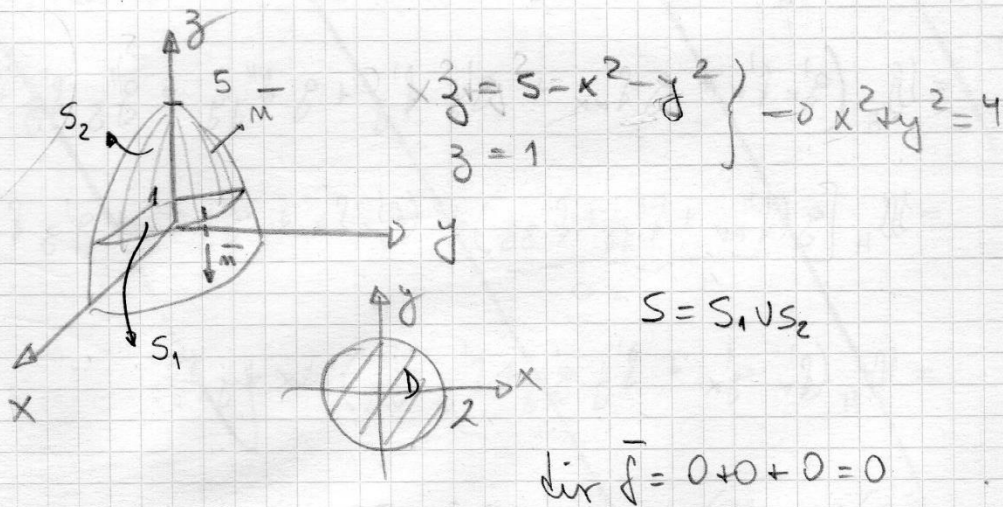
Calcule el flujo de $\vec{f}(x,y,z)=(g(y,z), h(x,z), 3x^2)$ a través de la superficie Σ abierta de ecuación $z=5-x^2-y^2$ con $z \geq 1$; suponga $\vec{f} \in C^1$. Indique gráficamente la orientación que eligió para Σ

$$32) \iint_S \vec{f} \cdot \vec{n} \, d\sigma = \iint_{S_1} \vec{f} \cdot \vec{n} \, d\sigma + \iint_{S_2} \vec{f} \cdot \vec{n} \, d\sigma \quad \wedge \quad \iint_S \vec{f} \cdot \vec{n} \, d\sigma = \iiint_V \operatorname{div} \vec{f} \, dx \, dy \, dz \Rightarrow$$

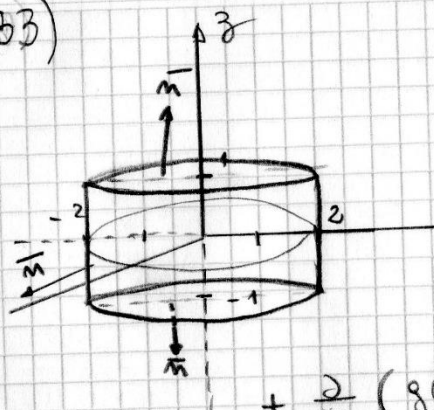
$$\Rightarrow \iint_{S_2} \vec{f} \cdot \vec{n} \, d\sigma = 0 - \int_0^{2\pi} \int_0^2 \underbrace{\downarrow}_{\vec{f} \cdot (0,0,-1)} (-3e^2 \cos^2 \theta) \, \rho = +3 \int_0^{2\pi} \cos^2 \theta \, d\theta$$

$$\cdot \left[\frac{\theta}{2} + \frac{1}{4} \sin 2\theta \right]_0^{2\pi} = +3 \cdot \frac{16^4}{4} \cdot \left[\frac{1}{2} \theta + \frac{1}{4} \sin 2\theta \right]_0^{2\pi} =$$

$$= +12 \left[\frac{1}{2} \cdot 2\pi + 0 - 0 - 0 \right] = +12\pi$$



33)



$$\iint_{\Sigma} \vec{f} \cdot \vec{n} \, d\sigma = \iiint_V \operatorname{div} \vec{f} \, dx \, dy \, dz = 0$$

$$\operatorname{div} \vec{f} = \frac{\partial (x + g(x,y))}{\partial x} + \frac{\partial (y + g(x,y))}{\partial y} +$$

$$+ \frac{\partial (g(x,y) - 2z)}{\partial z} =$$

$$= 1 + g'_x + 1 + g'_y - 2 = x + y + x - y = 2x$$

$$= \iiint_V 2x \, dx \, dy \, dz = 2 \int_{-1}^1 \int_0^{2\pi} \int_0^2 \rho (\rho \cos \theta) \, \rho =$$

$$= 2 \cdot 2 \left[-\sin \theta \right]_0^{2\pi} \left[\frac{\rho^3}{3} \right]_0^2 = 4 \cdot 0 \cdot [\] = 0$$

34)поле модале $\rightarrow$ campo con dir nulo

$$\text{dir } \vec{f} = 0 \Rightarrow \cancel{g(2xy)} + \overbrace{g'(2xy) \cdot 2y}^{\circ \circ \circ \circ} x + \cancel{g(2xy)} + \overbrace{g'(2xy) \cdot 2x}^{\circ \circ \circ \circ} y +$$

$$+ xy - \cancel{2g(2xy)} = 0$$

$$4xy g' + xy = 0 \Rightarrow xy(4g' + 1) = 0 \Rightarrow 4g' + 1 = 0$$

$$\Rightarrow g' = -\frac{1}{4} \Rightarrow \frac{dg}{dt} = -\frac{1}{4} \Rightarrow \int dg = \int -\frac{dt}{4} \Rightarrow$$

$$\Rightarrow g = -\frac{t}{4} + c$$

$$f(x, y, z) = \left(z + x \left(-\frac{2xy}{4} + c \right), y \left(-\frac{2xy}{4} + c \right), \right.$$

$$\left. zxy - 2z \left(-\frac{2xy}{4} + c \right) \right) = \left(z - \frac{x^2 y}{2} + cx, -\frac{xy^2}{2} + yc, \right.$$

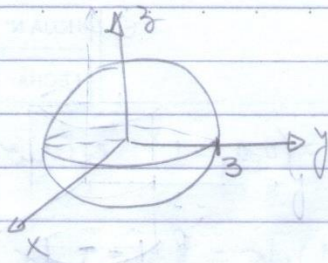
$$\left. \underbrace{zxy + \frac{xy^2}{2}}_{2xy \frac{z}{2}} - 2cz \right)$$

$$f(1, 1) = (3, 2, -3) \Rightarrow \begin{cases} 1 - \frac{1}{2} + c = 3 \\ -\frac{1}{2} + c = 2 \\ 2 - 2c = -3 \end{cases} \Rightarrow \begin{cases} c = 3 + \frac{1}{2} - 1 = \frac{5}{2} \\ -\frac{1}{2} + \frac{5}{2} = 2 \quad (V) \\ 2 - 2 \cdot \frac{5}{2} = -3 \quad (V) \end{cases}$$

$$g(t) = -\frac{t}{4} + \frac{5}{2}$$

$$\vec{f}(x, y, z) = \left(z - \frac{x^2 y}{2} + \frac{5x}{2}, -\frac{xy^2}{2} + \frac{5y}{2}, 2zxy - 5z \right)$$

35.)



$$\iint_S \vec{f} \cdot \vec{n} \, d\sigma = \iiint_V \operatorname{div} \vec{f} \, dx \, dy \, dz = (*)$$

$$\begin{aligned} \operatorname{div} \vec{f} &= \operatorname{div} [\nabla(g+h)] = \operatorname{div} (\nabla g + \nabla h) = \\ &= \operatorname{div} (\nabla g) + \operatorname{div} (\nabla h) = \nabla^2 g + \nabla^2 h = x^2 + y - z \end{aligned}$$

Laplaciano = 0 porque g es armónico

$$(*) = \iiint_V (x^2 + y - z) \, dx \, dy \, dz = \int_0^{2\pi} d\theta \int_0^{\pi} d\lambda \int_0^3 dp \cdot p^2 \operatorname{sen} \theta \cdot$$

$$\cdot (p^2 \cos^2 \theta \operatorname{sen}^3 \lambda + p \operatorname{sen} \theta \operatorname{sen} \lambda - p \cos \lambda) =$$

$$= \int_0^{2\pi} d\theta \int_0^{\pi} d\lambda \int_0^3 (p^4 \cos^2 \theta \operatorname{sen}^3 \lambda + p^3 \operatorname{sen} \theta \operatorname{sen}^2 \lambda - p^3 \cos \lambda \operatorname{sen} \lambda) =$$

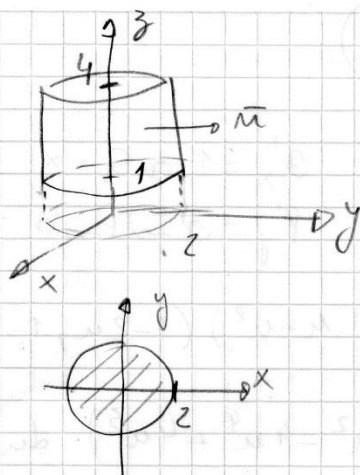
$$= \int_0^{2\pi} d\theta \int_0^{\pi} d\lambda \left[\frac{p^5}{5} \cos^2 \theta \operatorname{sen}^3 \lambda + \frac{p^4}{4} \operatorname{sen} \theta \operatorname{sen}^2 \lambda - \frac{p^4}{4} \cos \lambda \operatorname{sen} \lambda \Big|_0^3 \right] = \int_0^{2\pi} d\theta \int_0^{\pi} d\lambda \left(\frac{243}{5} \cos^2 \theta \operatorname{sen}^3 \lambda + \right.$$

$$\left. + \frac{81}{4} \operatorname{sen} \theta \operatorname{sen}^2 \lambda - \frac{81}{4} \cos \lambda \operatorname{sen} \lambda \right) =$$

$$= \int_0^{2\pi} d\theta \left[\frac{243}{5} \cos^2 \theta \underbrace{\left(\frac{4}{3} \right)}_{\int_0^{\pi} \operatorname{sen}^3 \lambda \, d\lambda} + \frac{81}{4} \operatorname{sen} \theta \cdot \underbrace{\frac{\pi}{2}}_{\int_0^{\pi} \operatorname{sen}^2 \lambda \, d\lambda} - \frac{81}{4} \underbrace{0}_{\int_0^{\pi} \cos \lambda \operatorname{sen} \lambda \, d\lambda} \right] =$$

$$= \left[\frac{972}{15} \underbrace{\int_0^{2\pi} \cos^2 \theta \, d\theta}_{\pi} + \frac{81\pi}{8} \underbrace{\int_0^{2\pi} \operatorname{sen} \theta \, d\theta}_0 \right] = \frac{972}{15} \pi = \frac{324}{5} \pi$$

36)



$$\begin{aligned}
 \iint_{\Sigma} \vec{F} \cdot \vec{n} \, d\sigma &= \iiint_V \operatorname{div} \vec{F} \, dx \, dy \, dz = \\
 &= \iiint_V (6a^3 + 6ab + b^2) \, dx \, dy \, dz = \\
 &= (6a^3 + 6ab + b^2) \int_0^{2\pi} d\phi \int_0^2 dp \cdot \\
 &\quad \cdot \int_1^4 dz =
 \end{aligned}$$

$$\begin{aligned}
 &= (6a^3 + 6ab + b^2) \cdot 2\pi \cdot \underbrace{\left[\frac{p^2}{2} \right]_0^2}_{=2} \cdot (4-1) = \\
 &\quad 12\pi
 \end{aligned}$$

$g(a, b) = (6a^3 + 6ab + b^2)$ para hallar extremos
 del flujo halle los extre-
 mos de g

$$\begin{cases} g'_a = 18a^2 + 6b = 0 & (1) \\ g'_b = 6a + 2b = 0 & (2) \end{cases}$$

De (2) $\rightarrow 2b = -6a \Rightarrow b = -3a$ reempl en (1):

$$\begin{aligned}
 18a^2 + 6(-3a) &= 0 \Rightarrow 18a^2 - 18a = 0 \Rightarrow 18a(a-1) = 0 \Rightarrow \\
 \Rightarrow a &= 0 \vee a = 1 & P_1 &= (0, 0) \\
 \downarrow b &= 0 & \downarrow b &= -3 & P_2 &= (1, -3)
 \end{aligned}$$

$$H = \begin{vmatrix} 36a & 6 \\ 6 & 2 \end{vmatrix} \quad H(0, 0) = \begin{vmatrix} 0 & 6 \\ 6 & 2 \end{vmatrix} < 0 \rightarrow (0, 0) \text{ silla}$$

$$H(1, -3) = \begin{vmatrix} 36 & 6 \\ 6 & 2 \end{vmatrix} > 0 \wedge g''_{aa}(1, -3) = 36 > 0 \rightarrow g(1, -3) \text{ es} \\
 \text{mín local}$$

$$\left. \begin{matrix} a = 1 \\ b = -3 \end{matrix} \right\} \text{mínimo local}$$